

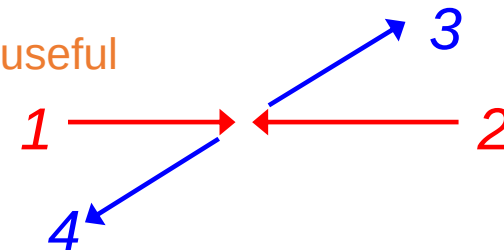
Particles and Nuclei 2

- 2/5/26
- Announcements:
- HW 2 Due a Wk from today 2/12/26
- No further reading for Tues may want to look ahead through slides

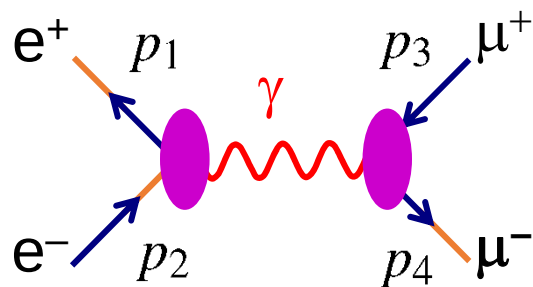
Mandelstam s , t and u

- ★ In particle scattering/annihilation there are three particularly useful Lorentz Invariant quantities: s , t and u

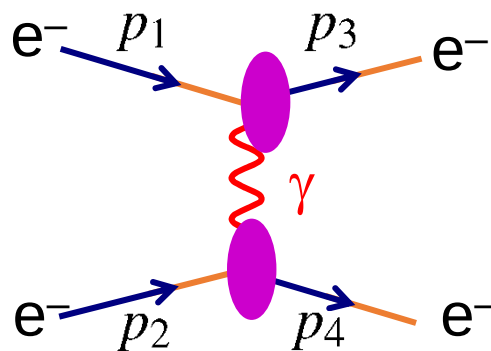
- ★ Consider the scattering process $1 + 2 \rightarrow 3 + 4$



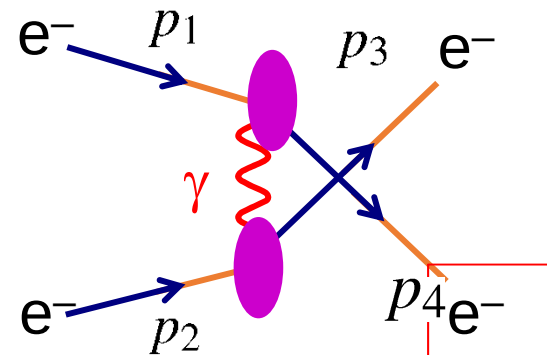
- ★ (Simple) Feynman diagrams can be categorised according to the four-momentum of the exchanged particle



s-channel



t-channel



u-channel

- Can define **three** kinematic variables: s , t and u from the following four vector scalar products (squared four-momentum of exchanged particle)

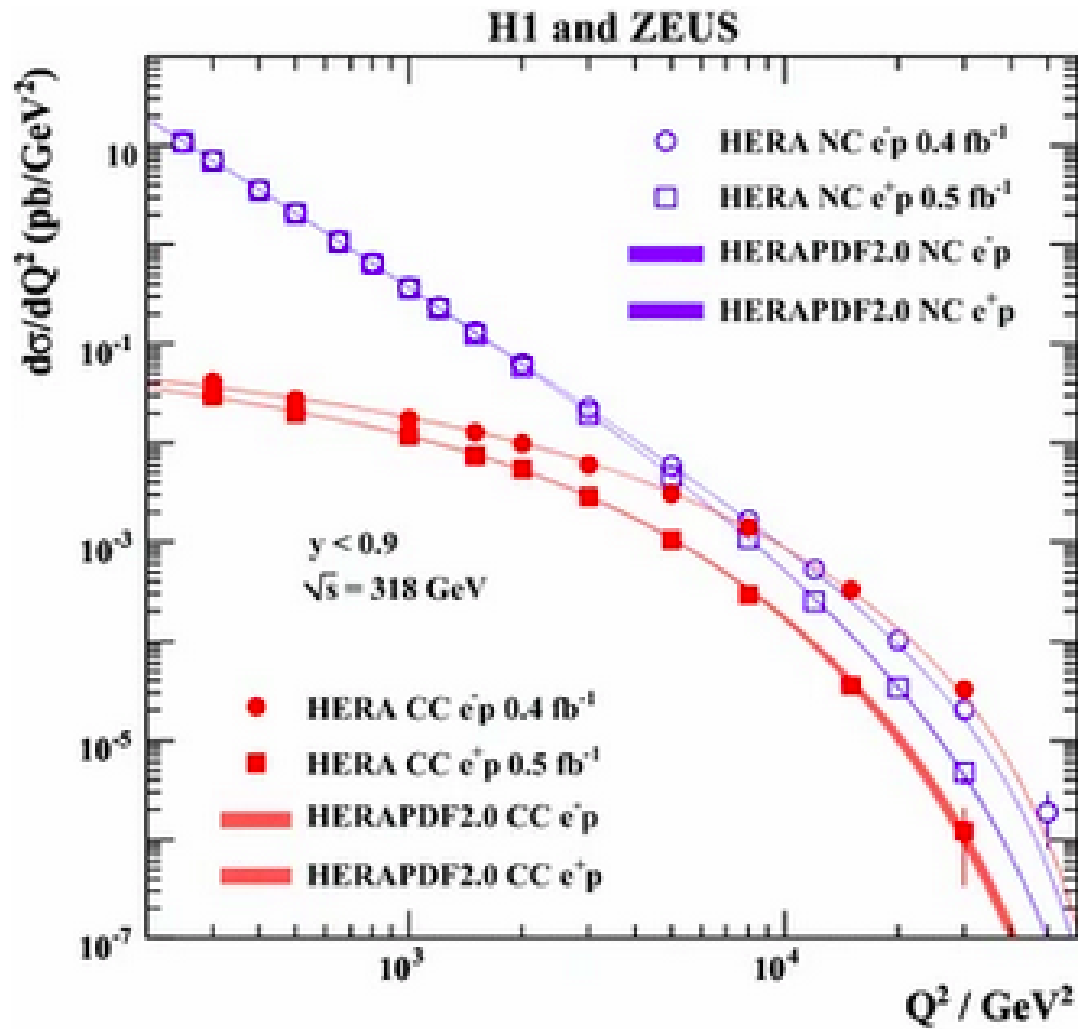
$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2$$

Other Dot Products/Invariant

- How about the invariant t

t, Q^2 kine limit

- Really high $\rightarrow$ Ebeam in Fixed Target



p1-p3

Dot products: $p^2 = \text{Mass}^2$ Most Important

- This relation is the most important 4-vector dot product
- $p \circ p = p^2 = m^2$
- Any momentum in ME bigger than m_{electron}
- ...and m_{pion} and m_{muon}
 - (thus can always ignore m / esp m^2)
- ...but not $m_{J/\Psi}$ (3 GeV) or sometimes not proton

List of Kinematics Topics

II. PLEASE REMEMBER

(masses are usually 0!)

A. $p = (E, \vec{p})$ is a 4-vector

B. Products of 4-vectors are invariant

C. $\frac{d^3p_{||}}{E}$ is an invariant, p_T is an invariant

$$D. \left. \begin{array}{l} p_{||} = m_T \sinh y \\ E = m_T \cosh y \end{array} \right\} \Leftrightarrow \left\{ \begin{array}{l} m_T = \sqrt{m^2 + p_T^2} \\ y = \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{array} \right.$$

$$E. \beta = p/E = \tanh y$$

$$F. E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$$

G. Lorentz transforms additive in rapidity

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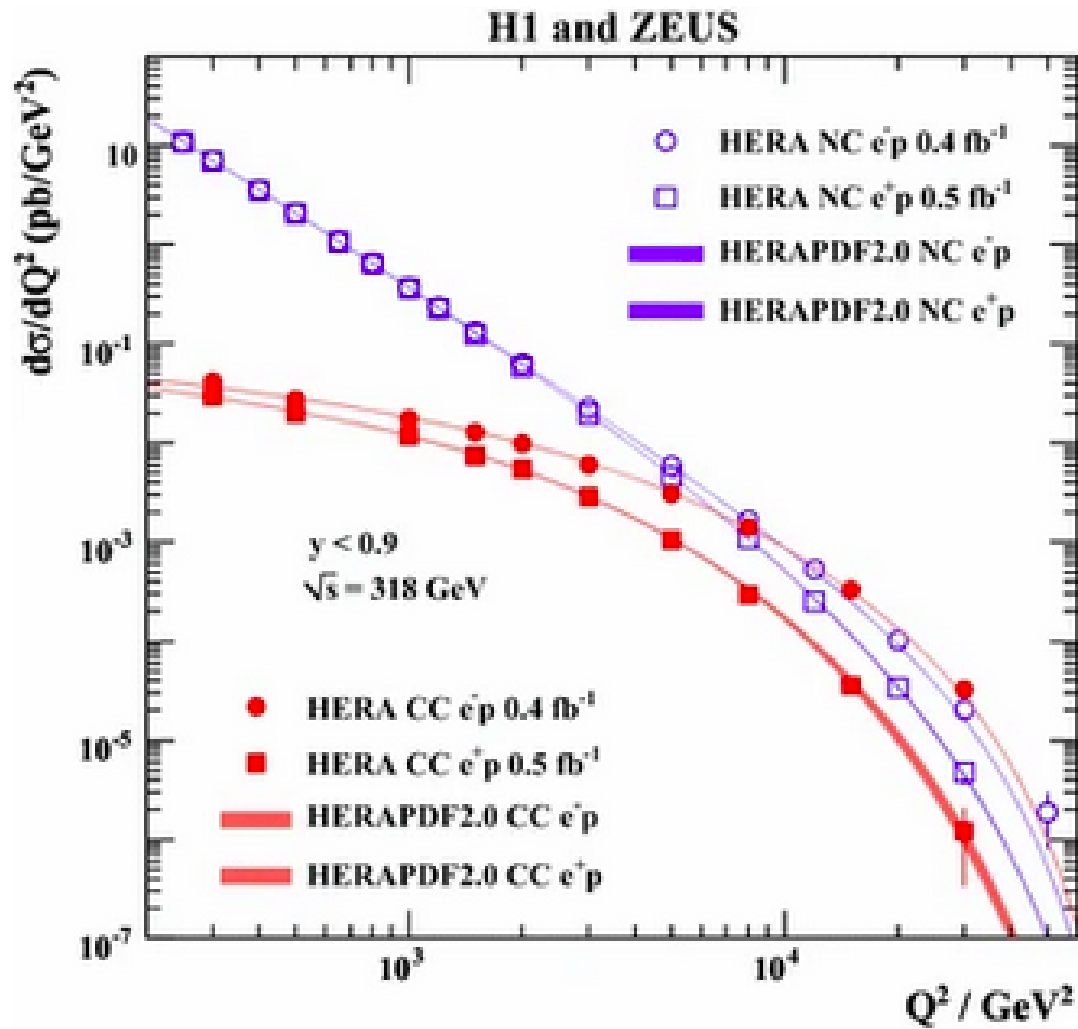
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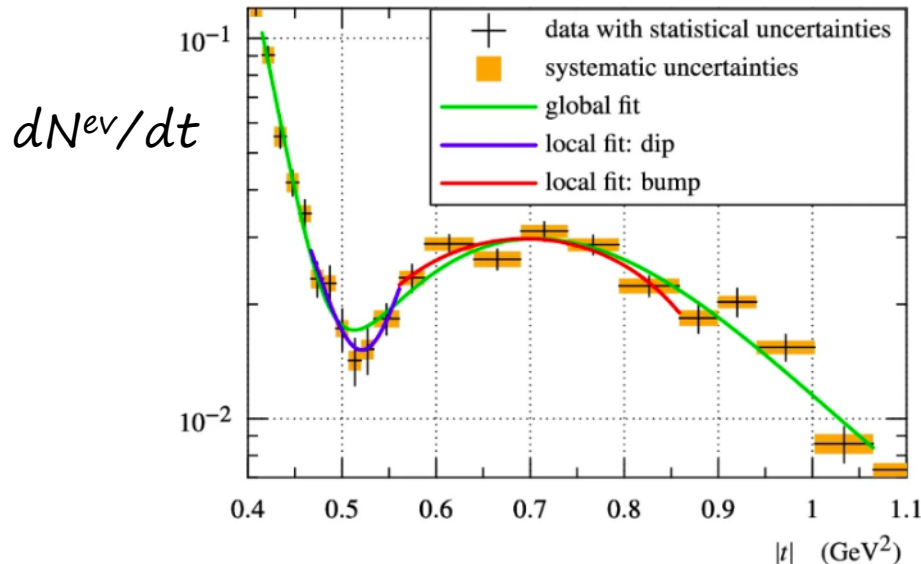
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p1-p3

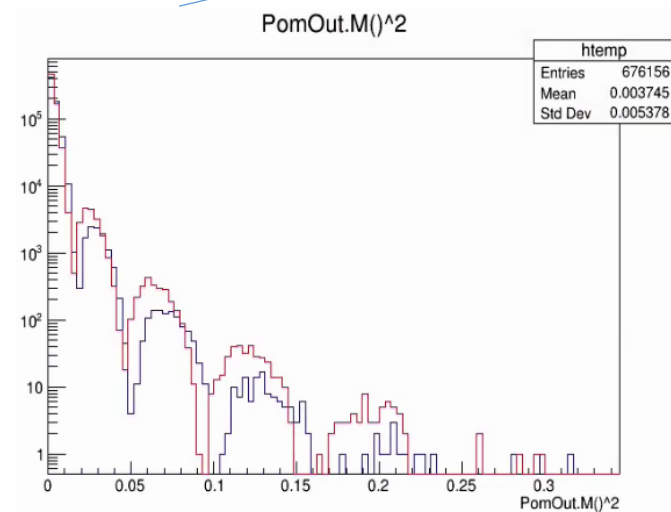
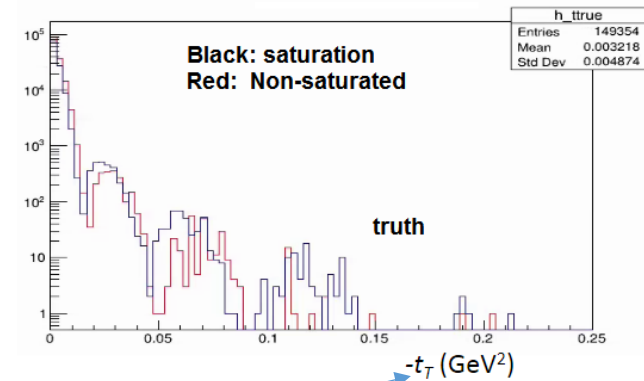
Proton “Image” Especially For Diffractive Scattering

- Plots like this “Form factor” at Jlab
- Measure size of hadrons/nuclei (diffraction)



Because of Fourier Transform
 $Q^2 \rightarrow t$

But this is very low $E \rightarrow$ the non-perturbative regime but because of the scale of the incoming (outgoing) E , makes treatment partially perturbative

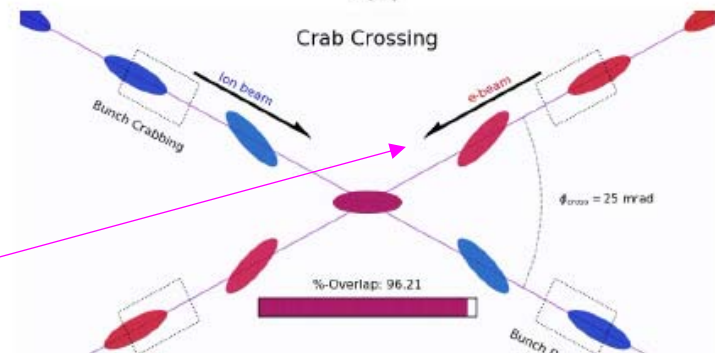
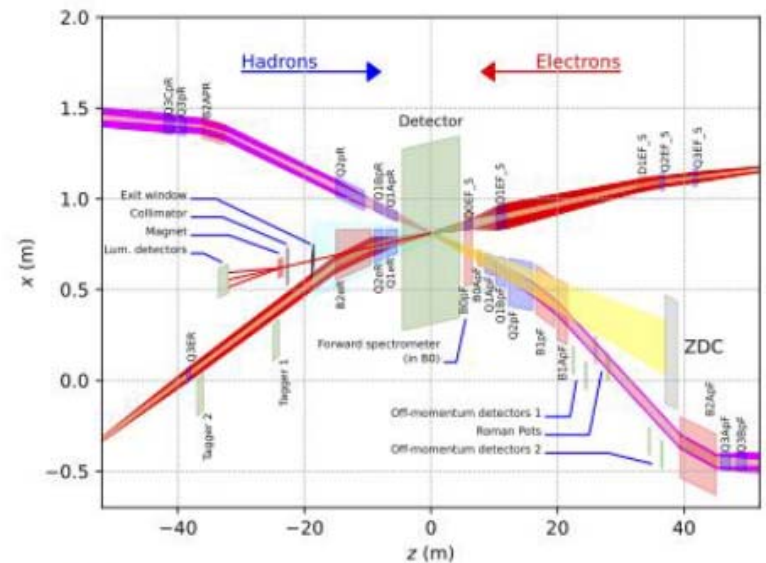


Addl Beam Boosts: EIC Crossing Angle

- Colliders can have some beam “crossing angle”
- The EIC will have about 10 times larger than RHIC

Leading EIC beam effects

- ▶ Unique accelerator with diverse beam effect direct impact exp.
 - Beam parameter in CDR section 3.1 and tables 3.3 to 3.5 and section 3.2
- ▶ 25-35 mrad beam crossing angle
- ▶ Angular beam divergence: $O(100\mu\text{rad})$
- ▶ Crab crossing (bunch-z dependent angle smear): $O(<100\mu\text{rad})$
- ▶ Beam energy spread $O(10^{-4})$
- ▶ Beam vertex spread from 10cm h-bunch collider with 1-cm e-bunch at finite crossing angle



This bunch rotate technique is called “crabbing”

Addl Beam Boosts: EIC Crossing Angle

- All accelerators have some beam “crossing angle”
- The EIC will have about 10 times larger than RHIC

Beam effect in ECCE analysis 1: two frames

- ▶ Head-on frame as used in most event generator and stored in HepMCEventMap is different from the Lab Frame as used in Detector design, Geant4 simulation and reconstruction
- ▶ In lab frame, electron is along $-z$ axis, i.e. along symmetric axis of exp. and no B-bending
- ▶ From head-on to lab frame, beam energy increase by
$$E_{Lab} = E_{HeadOn} / \cos(\text{crossing angle}/2)$$

Head-on frame (most event-gen, HepMC record)



Lab frame (Detector design, Geant4, reconstruction)



Going backwards -> "rotated" component(s)
boosted away in COM

Jin Huang

Addl Beam Boosts: EIC Crossing Angle

- All accelerators have some beam “crossing angle”
- The EIC will have about 10 times larger than RHIC

Beam effect in ECCE analysis 3: reconstruction

- ▶ Beam divergence ($O(100)\mu\text{rad}$) is not measured event-by-event, so in reconstruction we need to assume central beam four momentum with

$$E_{\text{Lab}} = E_{\text{HeadOn}} / \cos(\text{crossing angle}/2)$$

- Electron beam four vector: $(0, 0, -1, 1) * E_{e, \text{lab}}$
- Proton beam $Ip6, Ip8$: $(-\sin(25\text{mrad}), 0, \cos(25\text{mrad}), 1) * E_{p, \text{lab}}$
 $(+\sin(35\text{mrad}), 0, \cos(35\text{mrad}), 1) * E_{p, \text{lab}}$
- ▶ Lorentz invariant variable is reconstructed regardless frame, e.g. x-y-z-W-Q2-PhT
- ▶ Vectors and Lorentz variant need to be explicitly expressed with its frame, e.g. p_T , angle, pseudorapidity
- ▶ In sim we know e-by-e divergence. Truth Lorentz-rotation matrix between head-on and lab frame available at :
 - `CLHEP::HepLorentzRotation`
`PHHepMCGenEvent::get_LorentzRotation_EvtGen2Lab() const`
 - `CLHEP::HepLorentzRotation`
`PHHepMCGenEvent::get_LorentzRotation_Lab2EvtGen() const`
- ▶ in reco, we only know average beam vector. Therefore, lab to head-on boost in reco is
`TLorentzRotation().RotateY(12.5e-3).Boost( sin(12.5e-3), 0, 0)`

Head-on frame (most event-gen, HepMC record)



Lab frame (Detector design, Geant4, reconstruction)



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G. Lorentz transforms additive in rapidity

Exercise: for HW Invariance of $dp_{||}/E$

- **Exercise** To help prove above is invariant by directly transforming $dp_{||}$ and E
- Show that $dE/dp_{||} = p_{||}/E$
- Then...
- $dp_{||}' = \gamma(dp_{||} + \beta dE)$
- Cancel factors in ratio of $dp_{||}'/E'$

$$E = \sqrt{p^2 + m^2}$$

$$= \sqrt{p_x^2 + p_y^2 + p_z^2 + m^2}$$

$$= \sqrt{p_{||}^2 + p_T^2 + m^2}$$

$$\frac{dE}{dp_{||}} = \frac{p_{||}}{E}$$

$$p_{||}' = \gamma(p_{||} + \beta E)$$

$$E' = \gamma(E + \beta p_{||})$$

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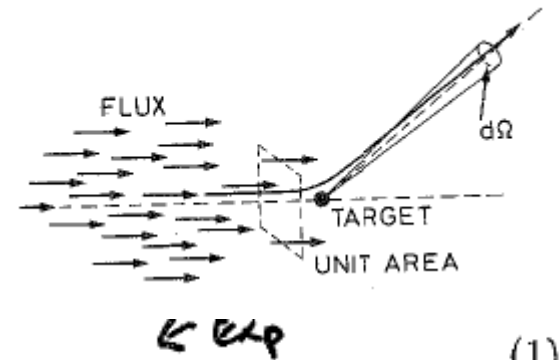
this is
9
Cross
Section
↓
Invariant
Cross Section

Cross Section x.s.

- $d\sigma/d\Omega$ might be the more fundamental quantity than σ
 - Think of this as x.s. as a fn of $\phi, \theta \rightarrow xs(\phi, \theta)$ “scattering amount in given direction”
 - usually just as a function of θ
- Total Cross section: $\sigma = \int d\sigma \rightarrow$ “fraction” (effective beam direction cross sectional area) of incident luminosity that experiences scattering $\rightarrow N_{scatt}^{total}$
- You are measuring the fraction that goes in the direction of various values of theta : $(\theta_i \rightarrow f_i)$

$$\frac{d\sigma}{d\Omega}(E, \Omega) = \frac{1}{F} \frac{dN_s}{d\Omega},$$

$$N_s = \frac{d\sigma}{d\Omega} N_0 n_T \Delta\Omega.$$



(1)

In general, $\frac{d\sigma}{d\Omega}$ depends on the type of projectile, the type of target, the energy of the projectile, and the scattering angle. Note also that the above formula is technically only exactly true in the limit that $\Delta\Omega \rightarrow 0$, due to the fact that $\frac{d\sigma}{d\Omega}$ depends on the scattering angle. The number of scattered particles N_s is directly proportional to N_0 , n_1 , and $\Delta\Omega$ as one intuitively expects – in this picture, the differential cross section is the proportionality constant. It is often said that the differential cross section “contains the physics”...

Cross Section $\rightarrow$ Just Counted N of something

- Cross Section (x.s.):

- Defined by equations like

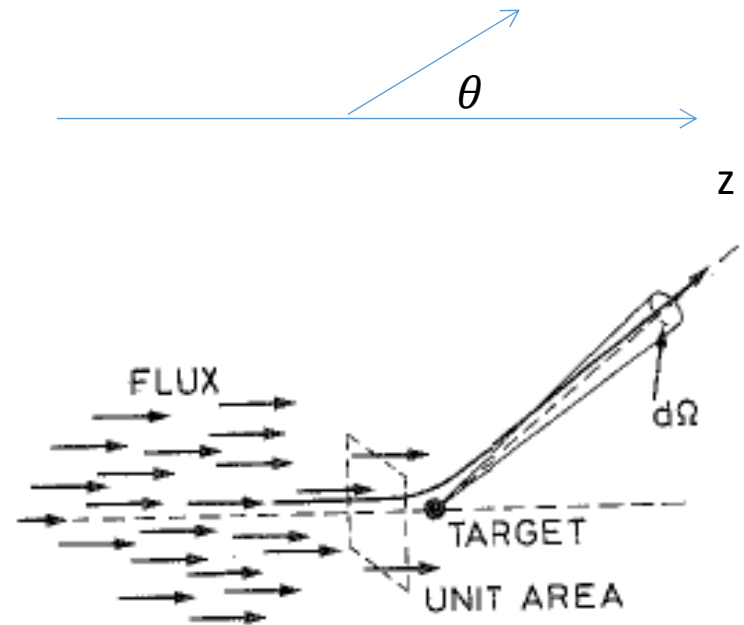
$$\text{Flux} * (\text{x.s.}) = \text{Scattering Yield}$$

- Luminosity / Flux F : particles per second per unit area in beam

- Often we integrate over time:
integrated luminosity / flux

- x.s. has units of area

Best to think of as weird normalization for scattered "yield"



$$\frac{d\sigma}{dx} \propto \frac{dn}{dx}$$

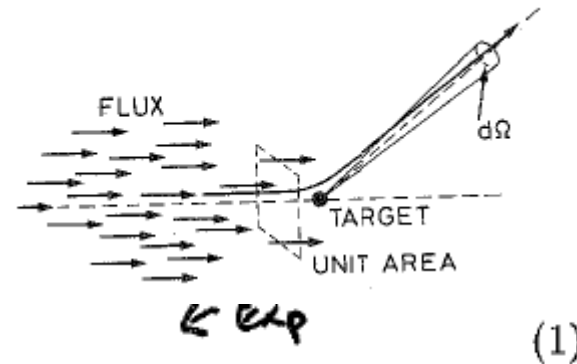
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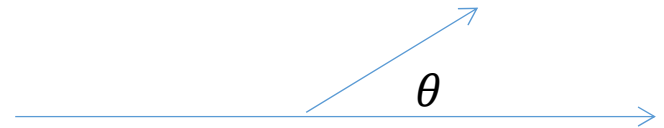
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Solid Angle

- A few “tricky” but actually simple quantities
- Solid angle : Ω serves double duty
 - 2 just short hand for angular direction $[\phi, \theta]$ (like a 2-D vector)
 - Solid angle itself is $\Delta\Omega$ or $d\Omega = d\phi d(\cos\theta) = \sin\theta d\theta d\phi$ (angular area $\perp \vec{\Omega}$)
- Differential yields: $dN/dx \rightarrow$ (think of as) Particle yield (number) as a function of x
 - $dN/dx = \int^{all y} dy dN/(dx dy)$
 - $\int d\phi dN/d\Omega = dN/d\theta$
 - $= \text{const } f(\theta) = 2\pi f(\theta)$
 - that's why Ruth formula only has theta in it

$d\Omega$ Sort of like calling this dA or dN/dA

$$d\sigma/d\Omega \propto dN_{scatt}/d\Omega = \frac{dN}{d\theta d\phi}$$



Why $\sin\theta$?

- how much area “in $d\phi$ ” depends on theta

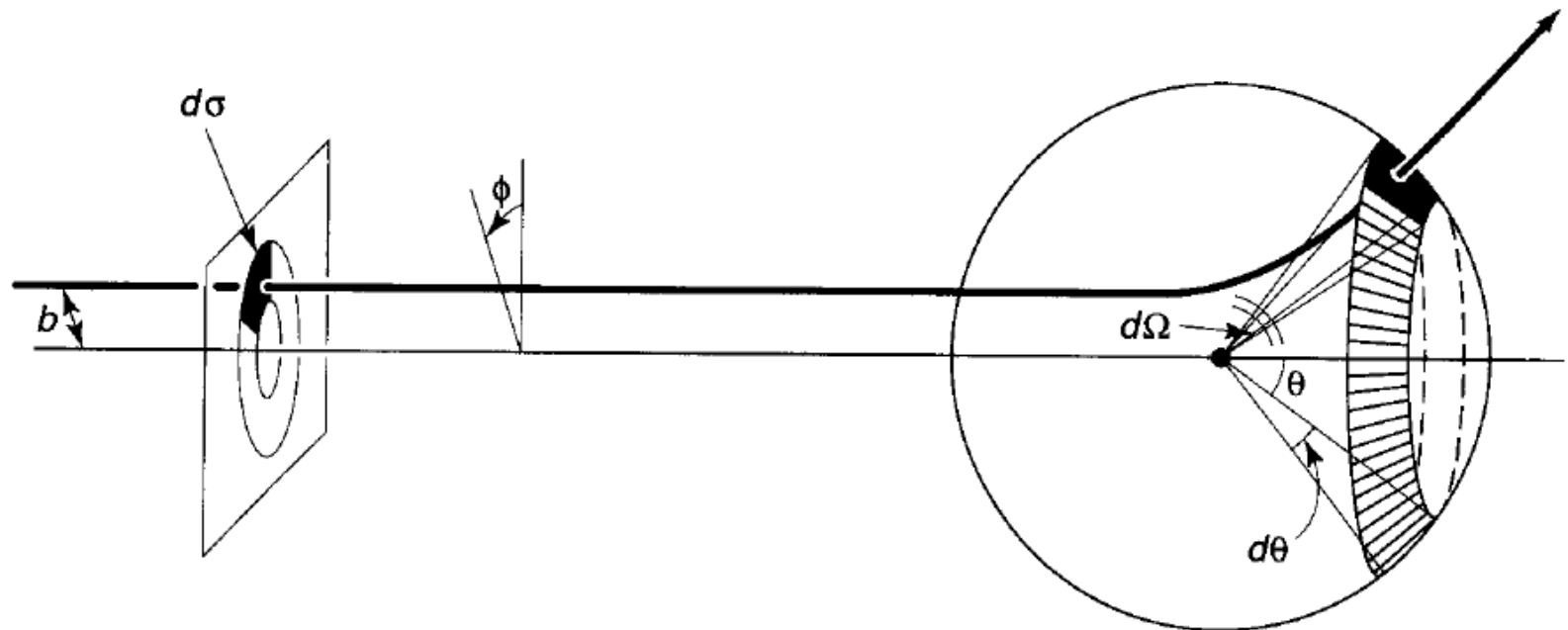


Figure 11.3: Particles incident in the area $d\sigma$ scatter into the solid angle $d\Omega$.

$$d\phi d(\cos\theta) = \sin\theta d\theta d\phi \rightarrow \text{just surface area on any sphere divided by } r^2$$

cos has same units as radians (because radians / sr are actually unitless units)!

Cross Section

- Cross Section (x.s.):

- Defined by equations like

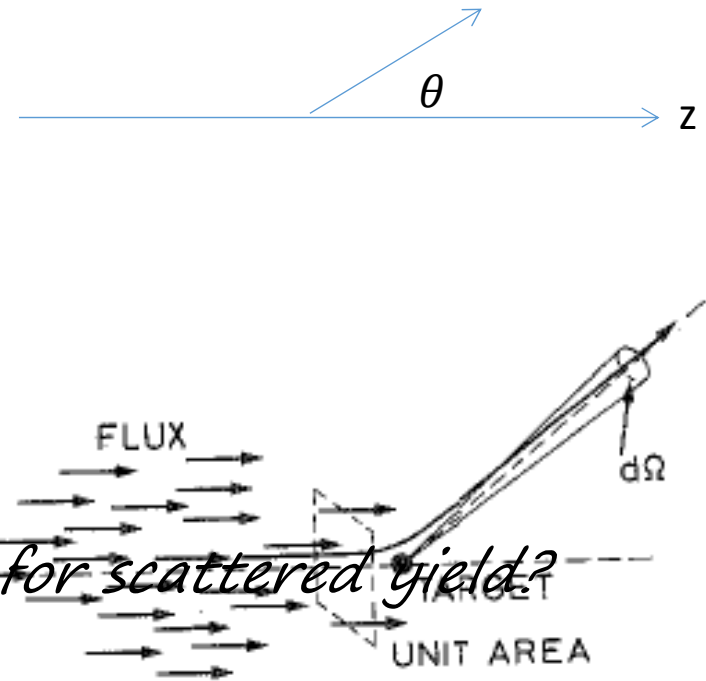
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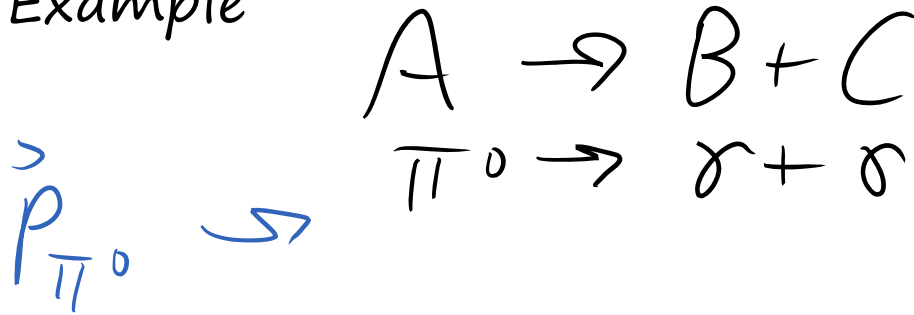
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G. Lorentz transforms additive in rapidity

show
cross
sections

Also can apply for DECAY distributions

- Example



$$\frac{dn}{d\theta_\gamma d\phi_\gamma} = ?$$

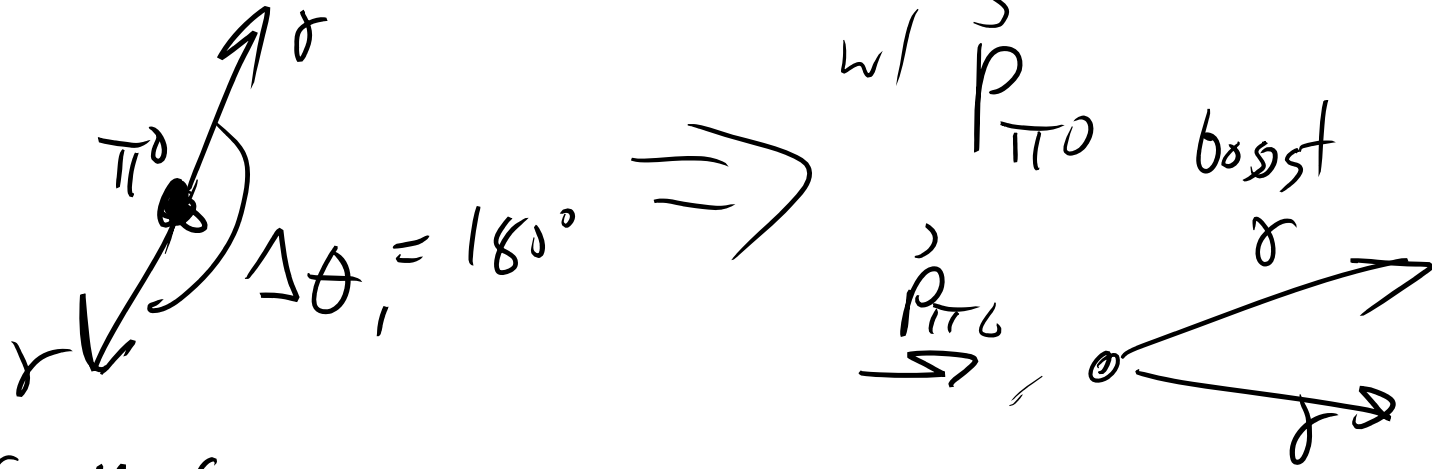
$$\frac{dn}{d\vec{p}^3/E} = \frac{dn}{dp_x dp_y (dp_z/E)}$$

$$\Downarrow \frac{dn_{\text{decay}}}{d\vec{p}_\gamma^3/E} = \text{same info}$$

How about boosted
decay distribution?
Group a relativistic $\vec{p}_{\pi^0}$

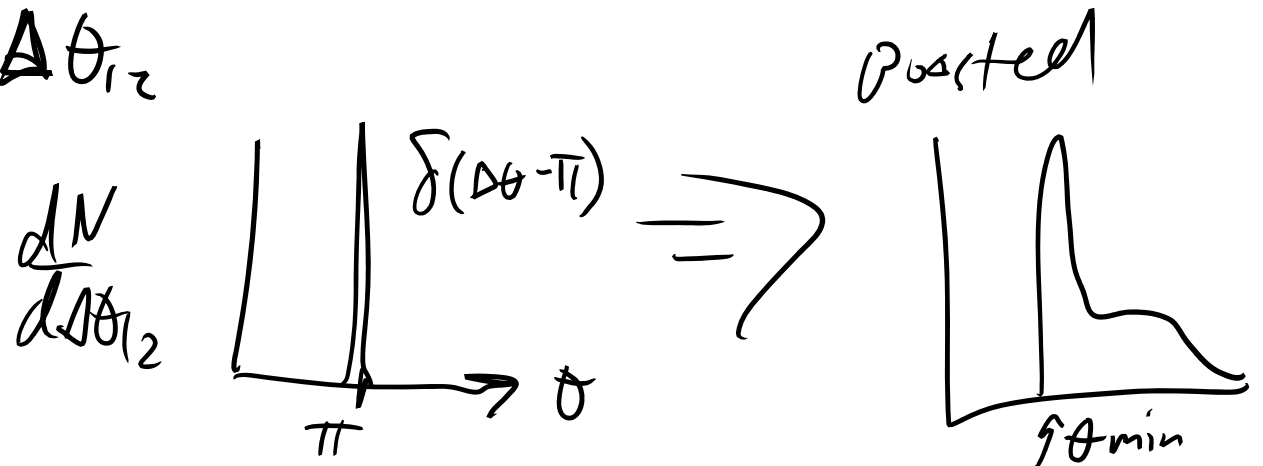
Next time

- Boosting distributions.
- Based on how the hard sphere scattering is handled



COM frame

open angle $\Delta\theta_{12}$



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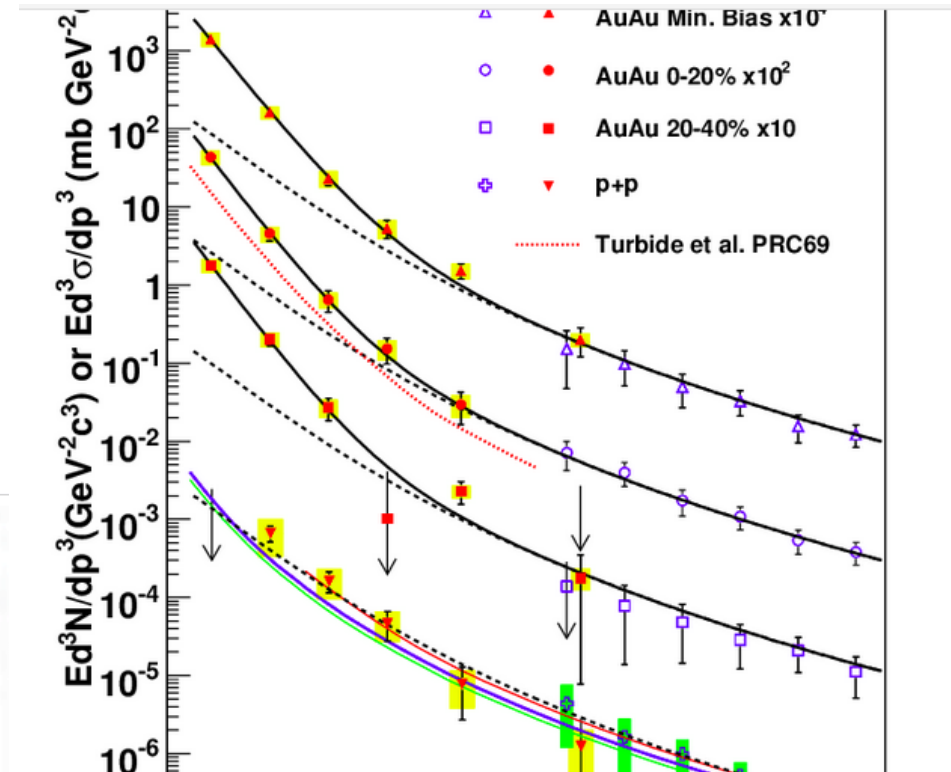
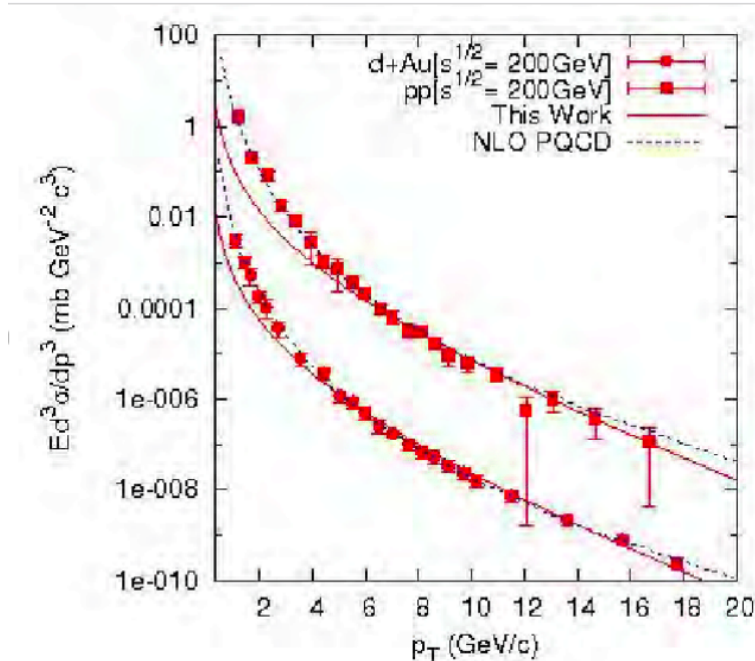
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show
cross
sections

Examples Published Inv Cross Sections.



A couple related relativistic effects in Field Theory

- The transition from single particle QM to relativistic Quantum Field Theory involves
 - 1) transition to multi-particle QM (can be done non-relativistic)
 - 2) several obvious generalizations of x, t, p, E into 4 vectors
- L/B Ch 4 covers the item 1
- L/B Ch 6 covers item 2

Standard Model (SM) – QFT's

- Field Expansion (again – “zoomed in”)

negative energy (anti-particle) states

negative sign in time
dependence attributed to
negative TIME not
negative energy (E/ω)

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left(a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^{\dagger} e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}} \right)$$

Act on generalized
number states $|0\rangle, |1\rangle,$
...

L/B

¹We retain the ‘particle-in-a-box’ normalization of Chapter 3. States are defined in a box of volume $\mathcal{V}$ (using periodic boundary conditions). For later convenience, we list some useful results using this normalization:

box’
are
sing
For
eful

4.1 Field operators

defined by

$$\hat{\psi}^\dagger(\mathbf{x}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} \hat{a}_{\mathbf{p}}^\dagger e^{-i\mathbf{p} \cdot \mathbf{x}}, \quad (4.7)$$

creates a particle at position $\mathbf{x}$, while $\hat{a}_{\mathbf{p}}^\dagger$ creates a particle in a state with three-momentum $\mathbf{p}$. Similarly, the operator $\hat{\psi}(\mathbf{x})$ defined by

$$\hat{\psi}(\mathbf{x}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} \hat{a}_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}}, \quad (4.8)$$

The inverse transform is

$$\psi_{\alpha}(\mathbf{x}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}} \tilde{\psi}_{\alpha}(\mathbf{p}), \quad (4.4)$$

since $\mathbf{p}$ takes discrete values. We also have

$$\int d^3x e^{-i\mathbf{p} \cdot \mathbf{x}} = \mathcal{V} \delta_{\mathbf{p},0} \quad (4.5)$$

A state $|\alpha\rangle$ can be described in position coordinates $\psi_{\alpha}(\mathbf{x}) = \langle \mathbf{x} | \alpha \rangle$ or momentum coordinates $\tilde{\psi}_{\alpha}(\mathbf{p}) = \langle \mathbf{p} | \alpha \rangle$, and since

$$\langle \mathbf{p} | \alpha \rangle = \int d^3x \langle \mathbf{p} | \mathbf{x} \rangle \langle \mathbf{x} | \alpha \rangle, \quad (4.1)$$

then the Fourier transform

$$\tilde{\psi}_{\alpha}(\mathbf{p}) = \frac{1}{\sqrt{\mathcal{V}}} \int d^3x e^{-i\mathbf{p} \cdot \mathbf{x}} \psi_{\alpha}(\mathbf{x}), \quad (4.2)$$

implies that

$$\langle \mathbf{p} | \mathbf{x} \rangle = \frac{1}{\sqrt{\mathcal{V}}} e^{-i\mathbf{p} \cdot \mathbf{x}}. \quad (4.3)$$

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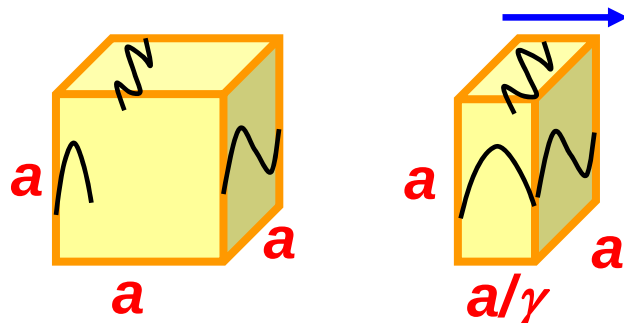
and

$$\frac{1}{\mathcal{V}} \sum_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{x}} = \delta^{(3)}(\mathbf{x}). \quad (4.6)$$

explains use of $\mathcal{V}$
factor (volume)
in WF norms
previously in
discussion of (dp/E)
and some “spatial
infinities”

Lorentz Invariant Phase Space “LIPS”

- In non-relativistic QM normalise to one particle/unit volume: $\int \psi^* \psi dV = 1$
- When considering relativistic effects, volume contracts by $\gamma = E/m$



- Particle density therefore increases by $\gamma = E/m$
 - ★ Conclude that a relativistic invariant wave-function normalisation needs to be proportional to E particles per unit volume
- Usual convention: **Normalise to $2E$ particles/unit volume** $\int \psi'^* \psi' dV = 2E$
- Previously used ψ normalised to 1 particle per unit volume $\int \psi^* \psi dV = 1$
- Hence $\psi' = (2E)^{1/2} \psi$ is normalised to $2E$ per unit volume
- Define **Lorentz Invariant Matrix Element**, M_{fi} , in terms of the wave-functions normalised to $2E$ particles per unit volume

$$M_{fi} = \langle \psi'_1 \cdot \psi'_2 \dots | \hat{H} | \dots \psi'_{n-1} \psi'_n \rangle = (2E_1 \cdot 2E_2 \cdot 2E_3 \dots 2E_n)^{1/2} T_{fi}$$

Example

- For the two body decay

$$i \rightarrow 1 + 2$$

$$\begin{aligned} M_{fi} &= \langle \psi'_1 \psi'_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i \cdot 2E_1 \cdot 2E_2)^{1/2} \langle \psi_1 \psi_2 | \hat{H}' | \psi_i \rangle \\ &= (2E_i \cdot 2E_1 \cdot 2E_2)^{1/2} T_{fi} \end{aligned}$$

- ★ Now expressing T_{fi} in terms of M_{fi} gives

$$\Gamma_{fi} = \frac{(2\pi)^4}{2E_i} \int |M_{fi}|^2 \delta(E_i - E_1 - E_2) \delta^3(\vec{p}_i - \vec{p}_1 - \vec{p}_2) \frac{d^3 \vec{p}_1}{(2\pi)^3 2E_1} \frac{d^3 \vec{p}_2}{(2\pi)^3 2E_2}$$

Note:

- M_{fi} uses relativistically normalised wave-functions. It is **Lorentz Invariant**
- $\frac{d^3 \vec{p}}{(2\pi)^3 2E}$ is the **Lorentz Invariant Phase Space** for each final state particle
the factor of $2E$ arises from the wave-function normalisation
(prove this in Question 2)
- This form of Γ_{fi} is simply a rearrangement of the original equation
but the **integral** is now **frame independent** (i.e. **L.I.**)
- Γ_{fi} is inversely proportional to E_i , the energy of the decaying particle. This is exactly what one would expect from time dilation ($E_i = \gamma m$).
- Energy and momentum conservation in the delta functions

Klein Gordon Equation

- Start w/ Klein Gordon Fields
- K.G. Governs scalar fields $\phi(x)$ (no spin)
- Could describe interactions of pions (meaning pion-pion) pretty straightforwardly
- Chiral EFT treatment of pion-pion – some technical limitation – goldstone boson etc...
- Higgs Particle!

$$(p_\mu)^2 = m^2$$

In operator form:

$$(\partial_\mu \partial^\mu + m^2) \phi = 0.$$

p^μ operator : Covariant Deriv

- why the greek index on the x is "up" (d/dx^ν) for the "down"-indexed ∂ operator ∂_ν

$$\partial_\nu = \frac{d}{dx^\nu}$$

$$\partial^\mu = \left(\frac{\partial}{\partial t}, -\frac{\partial}{\partial x^3} \right)$$

$$p_\mu \rightarrow i \partial_\mu$$

We can write this solution in four-vector form $\phi(x) = Ne^{-ip \cdot x}$. We'll call a wave with this sign combination in the exponential an **incoming** wave. Operating on it with momentum and energy operators yields

$$\begin{aligned} \hat{p}\phi(x, t) &= -i\hbar \nabla (Ne^{-i(\omega t - \mathbf{k} \cdot \mathbf{x})}) = \hbar \mathbf{k} \phi(x, t), \\ \hat{E}\phi(x, t) &= i\hbar \frac{\partial}{\partial t} (Ne^{-i(\omega t - \mathbf{k} \cdot \mathbf{x})}) = \hbar \omega \phi(x, t). \end{aligned} \quad (6.2)$$

plane waves

$$e^{i p \cdot x} = e^{-i p_\mu x^\mu} \quad \partial_\nu \psi = p_\nu \psi$$

First Reminder K.G. Notation

- Klein Gordon

$$\boxed{-\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} + \nabla^2 \psi = \left(\frac{mc}{\hbar} \right)^2 \psi}$$

Now instead of
 $E \propto p^2$

$$\partial_\mu \partial^\mu = \partial^\mu \partial_\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \equiv \square^2$$

$E^2 \propto p^2$

(E can be negative if
p is)

$$(\partial_\mu \partial^\mu + m^2) \phi = 0.$$

- Leads to negative probabilities AND negative energy solutions
- Negative energy solutions generally more OK $\rightarrow$ anti-particles
Dirac, (Proca? spin 1) has too

Dirac IDEA : Factorize "KG"

- Put plain wave solution into KG

$$\left(\partial_\mu \partial^\mu + m^2 \right) \phi = 0.$$

$$p^\mu = \begin{pmatrix} p^0, & \vec{0} \end{pmatrix}$$

$$p_\mu p^\mu - m^2 c^2 = 0 \Rightarrow (p^0)^2 - m^2 c^2 = 0 \Rightarrow \boxed{(p^0 + m)(p^0 - m) = 0}$$

This equation is easily satisfied by $p^0 = \pm mc$

- Idea for Dirac Eq mimic this idea make linear (only + or - will satisfy, choose +)

$$\partial_\mu \pm m = 0$$

First Reminder K.G. Notation

- Klein Gordon

$$\boxed{-\frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} + \nabla^2 \psi = \left(\frac{mc}{\hbar} \right)^2 \psi}$$

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(E can be negative if
p is)

$$(\partial_\mu \partial^\mu + m^2) \phi = 0.$$

- Leads to negative probabilities AND negative energy solutions
 - Negative energy solutions more OK $\rightarrow$ anti-particles Dirac, (“Proca” equation for spin 1,...) has too
 - Negative probability is a problem though $\rightarrow$ see next slide

Scattering Flux J

- Recall this quantity especially from QM:

$$J = \frac{i\hbar}{2m} (\psi \partial_x \psi^* - \psi^* \partial_x \psi) = \frac{\hbar}{2m} \text{Im}(\psi^* \partial_x \psi)$$

- Recall that it should satisfy “fluid dynamics” equation

$$\frac{d\rho(x)}{dt} = J$$

$$\rho(x) = \psi^* \psi$$

- We will also find it useful to define its relativistic counterpart
 - (the $2m$ may or may not be included — $\hbar = 1$)

$$j_\mu = i(\psi^* \partial_\mu \psi - \partial_\mu \psi^* \psi)$$

$$j^0 = \rho(x)$$

- The top form is derived thru Shrodinger, we may need changes that might not be obvious for relativistic versions like Klein-Gordon and Dirac eq — this one explicitly for K.G.

QFT: Proof of Needing Antiparticles #2 of 2

- Griffiths: K.G. : Leads to negative probabilities AND negative energy solutions
 - Negative energy solutions more OK $\rightarrow$ anti-particles Dirac, (Proca? spin 1 – see Lancaster/Bludell Massive Photon Fields section) has too
- Negative probability a problem – Griffiths: but not if we make $\Psi^*\Psi$ = charge density instead of number/counting probability
- That's not what ultimately happens though. See L/B
Instead, because the field equations have BOTH anti/particle terms Prob density/current FOR SCALAR (real) ϕ has equal ANTIPARTICLE/PARTICLE contributions

The resulting covariant probability current for the Klein-Gordon equation is then given by

$$j^\mu(x) = i \{ \phi^*(x) \partial^\mu \phi(x) - [\partial^\mu \phi^*(x)] \phi(x) \}, \quad (6.13)$$

which, as the notation suggests, is a four-vector. Substituting in our wave function $\phi(x) = Ne^{-ip \cdot x}$ gives the time-like component of the probability current as

The problem $\rightarrow j^0 = \rho = 2|N|^2 E. \quad (6.14)$

Since E can be positive or negative we can't interpret ρ as a probability density, since there's no such thing as negative probability! This seems to ring the death knell for the Klein-Gordon equation as a half-decent single-particle quantum mechanical equation.

j for $\phi(x)$ Antiparticles + Particles

- Solved by antiparticles added to the field operator

direction to the corresponding particle. A general solution to the Klein-Gordon equation with a particular positive energy now corresponds to a superposition of two states¹²

$$\phi(x) = \left[\begin{array}{c} \text{Incoming positive} \\ \text{energy particle} \\ \propto e^{-i(Et - \mathbf{p} \cdot \mathbf{x})} \end{array} \right] + \left[\begin{array}{c} \text{Outgoing positive} \\ \text{energy antiparticle} \\ \propto e^{+i(Et - \mathbf{p} \cdot \mathbf{x})} \end{array} \right]. \quad (6.19)$$

proof #1 of
antiparticles

This general solution is therefore not an inherently single-particle description, because we are forced to consider particles and antiparticles.

$$\phi(\mathbf{x}, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} \left(a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t + i\mathbf{p} \cdot \mathbf{x}} + a_{\mathbf{p}}^\dagger e^{i\omega_{\mathbf{p}} t - i\mathbf{p} \cdot \mathbf{x}} \right)$$

L/B also initially explains this with CHARGED antiparticles : j is $(-)$ for neg charge but later explains that for the scalar real $\phi(x)$, $\phi(x)$'s are their own antiparticles ...implying that their "current"/density is always net zero, with a positive/negative cancelation

For more realistic non -net zero prob currents,
use COMPLEX scalar fields ϕ , ϕ^* [or. DOUBLET ϕ_1 , ϕ_2]

Complex Scalar Fields – QFT

- Complex Scalar fields [not Dirac/Fermion fields] are the QFT prototypical example of making a EM-like charge :
- “U(1)” charge -- the kind that can just cancel/annihilate like EM charge
- L/B “Complex Scalar field theory section”

Step I: We start by writing down the Lagrangian we want to quantize.
The Lagrangian for complex scalar field theory is

$$\mathcal{L} = \partial^\mu \psi^\dagger(x) \partial_\mu \psi(x) - m^2 \psi^\dagger(x) \psi(x). \quad (12.1)$$

Step II: Identify the momentum density and write down the Hamiltonian. Each component² σ of the field (i.e. $\sigma = \psi$ and $\psi^\dagger$) has a different

Lagrangian

$$+ \frac{1}{2} [\partial_\mu \phi_2(x)]^2 - \frac{1}{2} m^2 [\phi_2(x)]^2$$

We noticed that we can make a transformation that simplified this Lagrangian. We defined

$$\begin{aligned} \psi &= \frac{1}{\sqrt{2}} [\phi_1(x) + i\phi_2(x)], \\ \psi^\dagger &= \frac{1}{\sqrt{2}} [\phi_1(x) - i\phi_2(x)], \end{aligned}$$

similar to two interacting real $\phi_1 \phi_2$'s – Griff's Ch 6 ABC theory (ABB see next slide) or 2 component vector fields

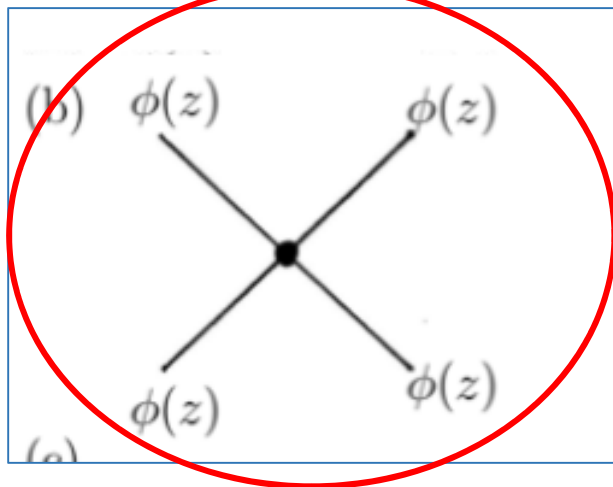
in those cases, though doesn't have this “standard” kind of “charge” “charge” here is a true conserved +/- derived from Noethers

Also notice NOT 2 complex fields → (only 2 “components”, not 4 free “components”)

Interactions

Chapter 4/5 actually discusses these interactions already and even introduces the Feynman diagram to represent it

These are perturbations on the free particle Hamiltonian



19.1 Meet some interactions

Example 19.1

L/B

(ii) The simplest self-interaction of fields is ϕ^4 (say 'phi-fourth' or 'phi-four') theory described by a Hamiltonian

$$\mathcal{H}_1(z) = \frac{\lambda}{4!} \phi(z)^4. \quad (19.3)$$

ϕ^4 self interactions are prototype for both QFT books for describing Feynman diagrams unfortunately

This is what is used for pion+pion scattering in Chiral EFT

Rapidity y and η (Pseudo-)

- Nucl lunch
- Two almost identical quantities that are useful partially because they both exist –
- use/derive properties of one applied to the other
- Note, it may seem from slides rapidity more related to Heavy Ion physics, but it is actually more of a COLLIDER thing
 - LHC [p+p] / HERA / Tevatron/ EIC communities all use this extensively!!!
 - Not sure why it's not covered in books, but I suspect they were ~all written by people who did much of their work before the prevalence of these highest energy colliders – and b.c. of the desire to connect to normal Quantum scattering coverage
 - Maybe also, only at these higher energy colliders did people start to explore more forward/backward angle physics –because it was inaccessible in fixed target
 - Also many initially important collider cross sections are strongly peaked at 90 degrees from beam where rapidity is not as important?

Start w/ Pseudo-rapidity

- Very closely related quantity : η -- Pseudo rapidity

In experimental particle physics, **pseudorapidity**, a coordinate describing the angle of a particle relative to the beam axis as

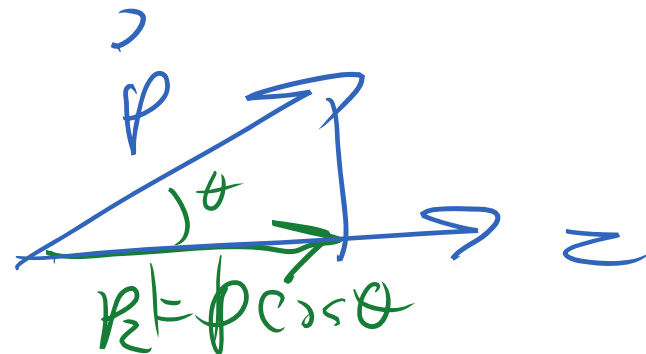
$$\eta \equiv -\ln \left[\tan \left(\frac{\theta}{2} \right) \right],$$

where θ is the angle between the particle three-momentum and the direction of the beam axis.^[1] Inversely,

$$\theta = 2 \arctan(e^{-\eta}).$$

As a function of three-momentum $\mathbf{p}$, pseudorapidity

$$\eta = \frac{1}{2} \ln \left(\frac{|\mathbf{p}| + p_L}{|\mathbf{p}| - p_L} \right) = \operatorname{artanh} \left(\frac{p_L}{|\mathbf{p}|} \right),$$



$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

- For massless particles (γ , $\sim \nu$, $\sim e \dots$) $y = \eta$ exactly
- Even for $\sim$ all massive particles they y, η almost identical except very VERY far forward and backward

Pseudo-rapidity

- Very closely related quantity : η -- Pseudo rapidity

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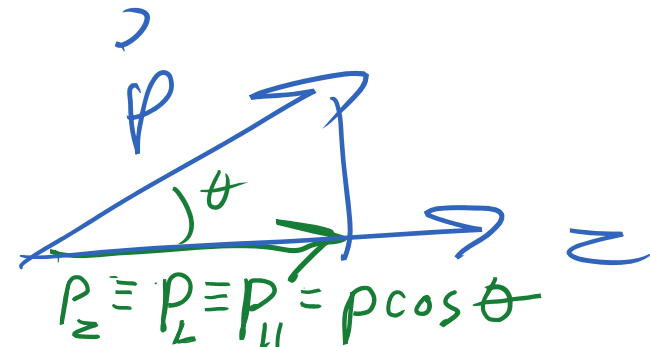
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$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

Why same? $\Rightarrow \frac{1}{2} \ln \left(\frac{\frac{1}{p}(p + p_L)}{\frac{1}{p}(p - p_L)} \right)$ ← Same

- For massless particles (γ , $\sim \nu$, $\sim e$...) $y = \eta$ exactly [see later]
- Even for $\sim$ all massive particles they y, η almost identical except very VERY far forward and backward

Equivalent Definitions of Pseudorapidity η

- z means z axis in coordinate system with z in beam direction. Theta, θ , is the polar angle in spherical coordinates (r, θ, ϕ) . $p = |\vec{p}|$ is the magnitude of the 3-mom

$$\eta = \frac{1}{2} \ln \left(\frac{p + p_z}{p - p_z} \right)$$

$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

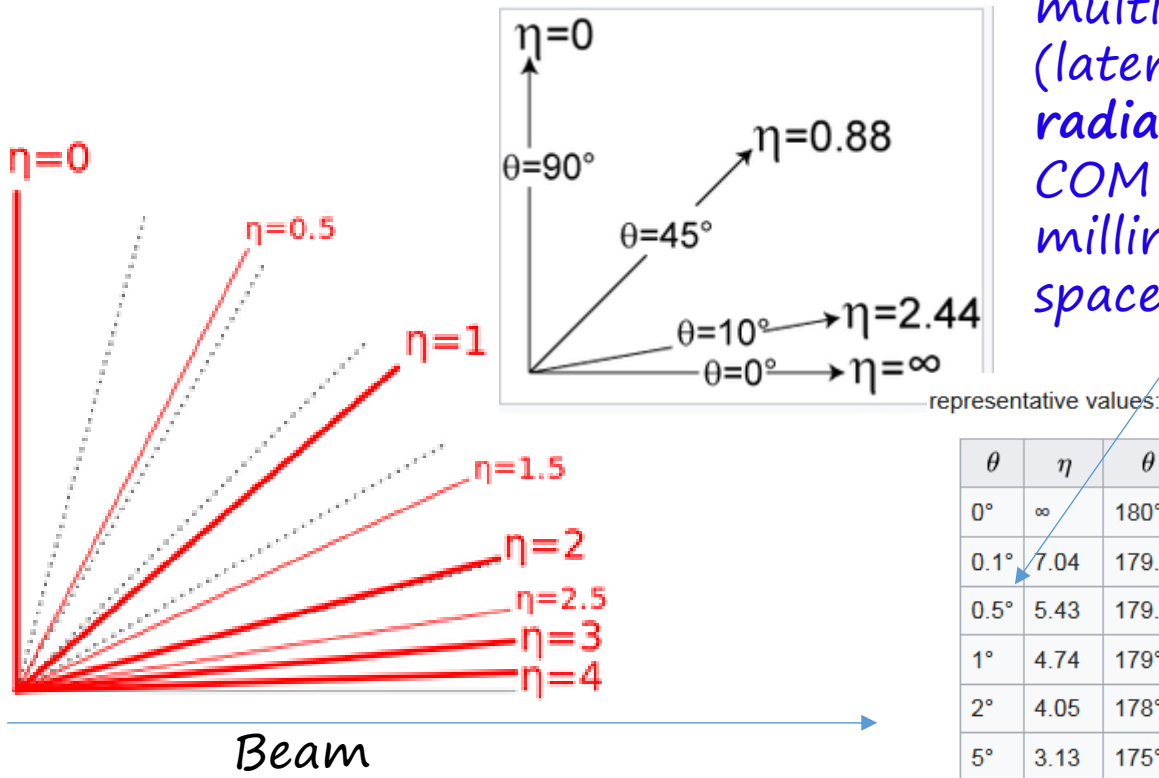
$$\eta = -\frac{1}{2} \ln \left(\tan \frac{\theta}{2} \right) \quad \eta = \frac{1}{2} \ln \left(\cot \frac{\theta}{2} \right)$$

$$\eta = \operatorname{arctanh}(p_z / p)$$

$$\eta \leftrightarrow \theta$$

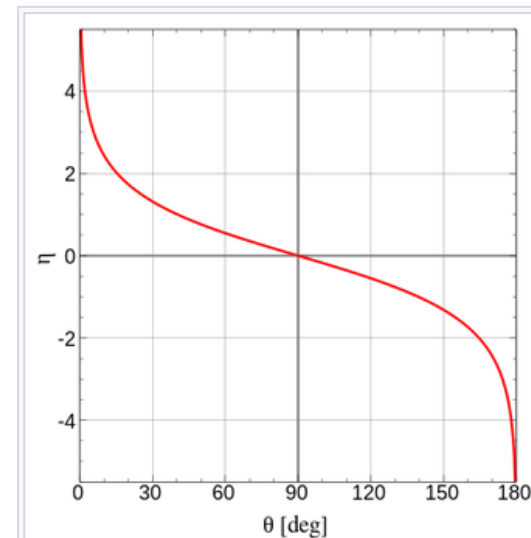
- Pseudo rapidity allows an easier mapping to θ and shows us why this version of θ is so much better
- (From Wikipedia:pseudorapidity):

More than simply " $\ln\theta$ " : Close to beam, boost scrunches up multiple units of rapidity (later slides: = multiple radians [1 rad = 60 deg] of COM angular space!) in milliradians of lab angular space!!!



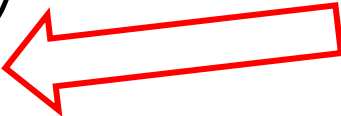
θ	η	θ	η
0°	∞	180°	$-\infty$
0.1°	7.04	179.9°	-7.04
0.5°	5.43	179.5°	-5.43
1°	4.74	179°	-4.74
2°	4.05	178°	-4.05
5°	3.13	175°	-3.13
10°	2.44	170°	-2.44
20°	1.74	160°	-1.74

150°	-1.32
------	-------



Particle Physics Rapidity y

- Note there is another slightly different definition of rapidity y (a different, but related, quantity) in textbooks. The one we are interested in is the “particle physics” / collider version
- Experimentally, for this more usual particle physics kind, we typically **ONLY** consider boosts (& ll direction as being) along the (common) **beam direction p_z -- common to all particles**
 - where as the other version is in each particle's direction
- Our version will be defined similarly to the pseudo-rap:

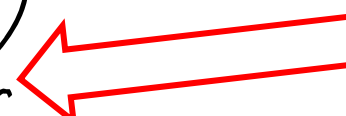
$$y = \frac{1}{2} \ln \left(\frac{E + p_{z/\text{beam}}}{E - p_{z/\text{beam}}} \right)$$


where z is the “global” coordinate z along the beam direction

- **Example :** what will y be for a particle moving perpendicular to beam (ie with $\theta = 90$ deg)
 - In that case $p_z = 0$
 - $y = \frac{1}{2} (\ln (E + 0/E-0)) = \frac{1}{2} \ln 1 = 0$

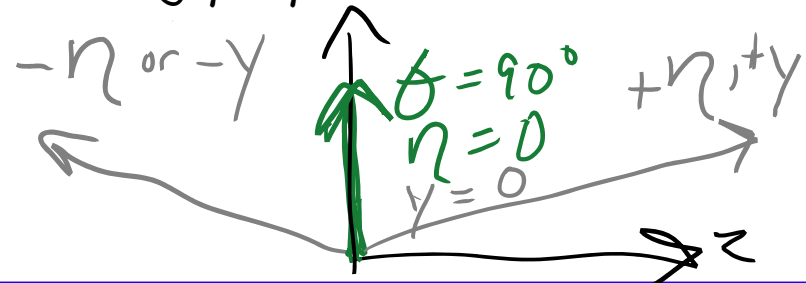
Particle Physics Rapidity y

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$$y = \frac{1}{2} \ln \left(\frac{E + p_{z/\text{beam}}}{E - p_{z/\text{beam}}} \right)$$


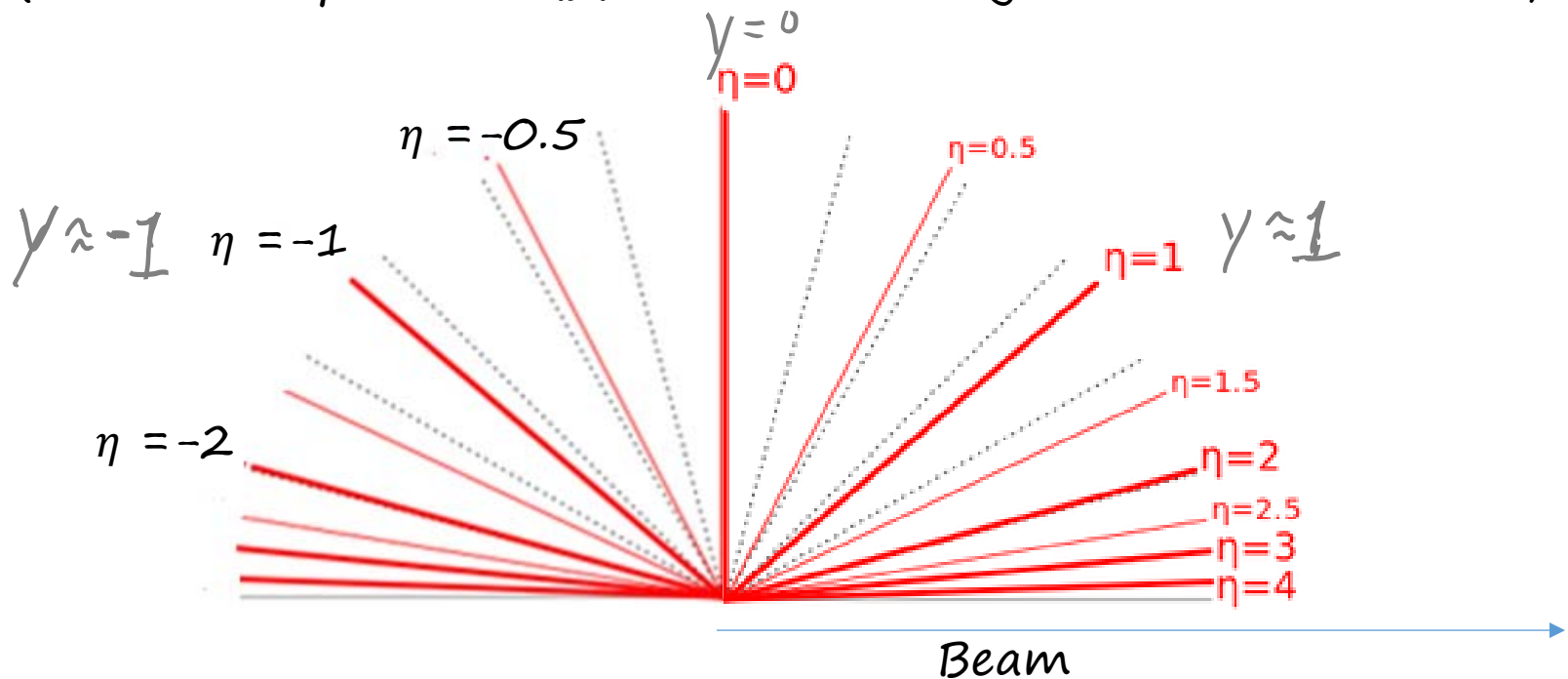
where z is the "global" coordinate z along the beam direction
 Compared to η the magnitude of 3-mom p is replaced by the Energy E

- **Example:** what will y be for a particle moving perpendicular to beam (ie with $\theta = 90^\circ$)
 - In that case $p_z = 0$
 - $y = \frac{1}{2} (\ln (E + 0/E - 0)) = \frac{1}{2} \ln 1 = 0$.
 - also ~same for η



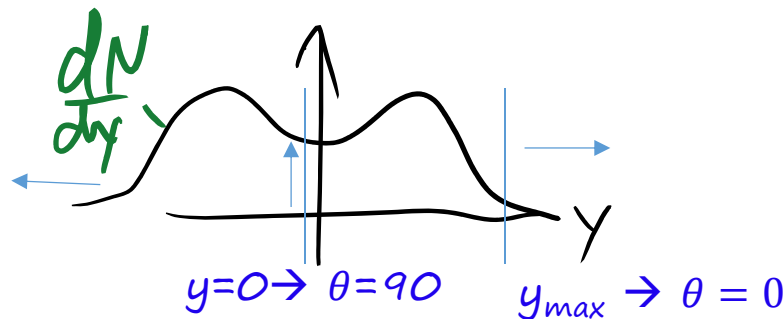
y, η – negative/positive

- So η and y also go from negative to positive symmetrically
- Technically η always can range from $-\infty$ to ∞ ...
- But as we will see, values we actually CARE ABOUT will be limited to $[-n, +n]$ where n is a integer $< \sim 10$...
- [because y (rather than η) will typically be limited to that !]
 - (for massive particles, $|y|$ does not “diverge” ie never reaches ∞)



Important Q: What is y ? Same as η !

- What should we think of **rapidity** as physically?
 - Best to think of it as same as pseudorapidity ! (more “fundamental” thing?)
- Some describe it as like a velocity (because it transforms like classical velocity for Galilean velocity addition as we’ll see next)
- I suggest thinking of it better as just a **direction** more specifically the “ z ” component of a particles direction: (ie like the θ angle—but better relativistically) – ie like η !
 - This is partially because of the more specific way we define it in particle physics w.r.t the beam axis (more on this soon)
 - and also especially because of it’s similarity to pseudo-rapidity (see next and later slides)



Another form of Rapidity Def

- Consider the following “shorter” definition which starts with our previous definition

$$y = \frac{1}{2} \ln \left(\frac{E + p_{\parallel}}{E - p_{\parallel}} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1 + \beta_{\parallel}}{1 - \beta_{\parallel}} \right)$$

special “ β ” (“ $\beta_{\parallel}$ ”)
--equals actual
boost β when p is
completely in “ $\parallel$ ”
direction (for us
the z -direction)

- In both of these versions, it might seem like we could define the direction of the z axis in any direction, but in fact we ONLY consider “parallel” and “transverse” with respect to the BEAM direction. For us, therefore the $p_{\parallel}$ is explicitly the same as p_z , i.e. $p_{\parallel} \equiv p_z$.

Actual Rapidity y Definitions:

- This slide contains several equivalent formulae to define y
- We define y very similiarly to η but with E rather than $|p|$

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right)$$

$$y = \frac{1}{2} \ln \left(\frac{1 + p_z/E}{1 - p_z/E} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta_z}{1 - \beta_z} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta \cos \theta}{1 - \beta \cos \theta} \right)$$

Because of the above it is also true that for example:

$$y = \operatorname{arctanh}(\beta_z) \quad \text{or} \quad \tanh(y) = \beta \cos \theta$$

this differs from a different but somewhat common definition for rapidity in textbooks of $\tanh y = \beta$ which would only be true for our version for the beam particles themselves (or those with $\theta = 0/180\text{deg}$)

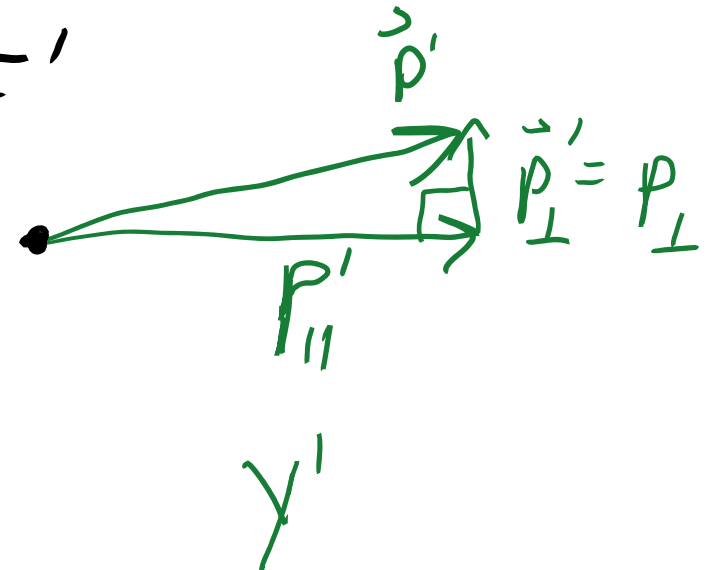
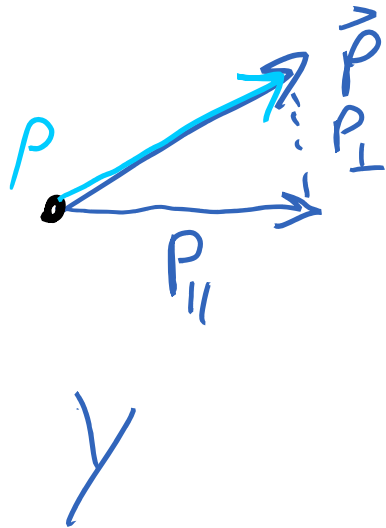
Rapidity

- Consider the following (why? – see next slides)

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

obviously this is either a special " β " (" $\beta_{||}$ ") or is only true when p is completely in " $||$ " direction on z -direction

- Consider y' of a particle in an arbitrary other boosted frame F' (e.g. in frame of another moving particle F' (for us Beam particles in the z -direction) vs rest frame
- To be clearer what is meant – particle is P



Even though diagonal

• df

Rapidity

- Consider the following (why? - see next slides)

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta_{||}}{1 - \beta_{||}} \right)$$

note this is regular β when p is completely in "||" direction on z-direction

- Consider y' of a particle in the boosted frame F' (of the beam particles) vs rest frame using

$$\begin{aligned} p'_{||} &= \gamma(p_{||} + \beta E) \\ E' &= \gamma(E + \beta p_{||}) \end{aligned} \Rightarrow y' = \frac{1}{2} \ln \left(\frac{E' + p'_{||}}{E' - p'_{||}} \right) = \frac{1}{2} \ln \left(\frac{\cancel{\gamma} E + p + \beta(E + p)}{\cancel{\gamma} (E - p) - \beta(E - p)} \right)$$

$$= \frac{1}{2} \ln \left[\left(\frac{E + p_{||}}{E - p_{||}} \right) \left(\frac{1 + \beta_F}{1 - \beta_F} \right) \right] = y + y_{\beta_F}$$

$\frac{1}{2} \ln(\downarrow) + \frac{1}{2} \ln(\downarrow)$

β is boost of F' itself ($\neq p/E$ of particle P , rather $= p_{||F}/E_F$)
 $y_{\beta} \rightarrow$ rap of $F \equiv y_F$
 & notice therefore it's satisfying the above caveat in red

Again as noted in the colored comments, since we are boosting into the frame F' , the beam particles, which have $\cos\theta = 1$ the actual boost β becomes the β_F that defines the y_{β_F} . We call y_{β_F} the **BEAM RAPIDITY**

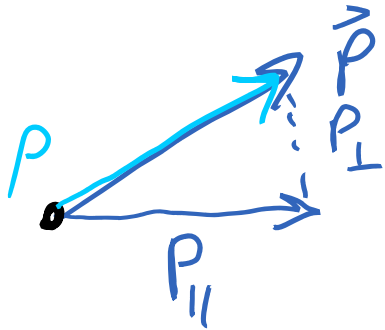
Beam Rapidity Visualization

- Consider the following

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta_{||}}{1 - \beta_{||}} \right)$$

$$y' = y + y_{\text{Beam}} \rightarrow "y_{BF}"$$

- Visually :



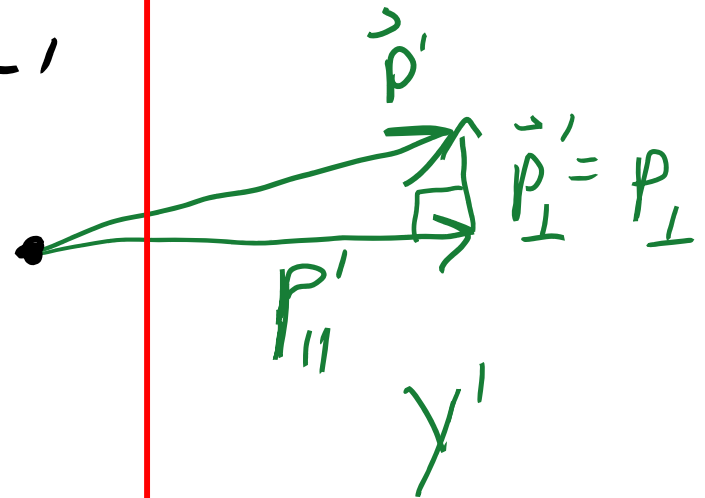
y



$$\vec{p}_{||F} = \vec{p}_F$$

$$\beta_{||} = \beta$$

$$y_{BF}$$



• df

Boosted Rapidity - Implication

- Back to our derivation

$$y = \frac{1}{2} \ln \left(\frac{E + p_{||}}{E - p_{||}} \right) = \frac{1}{2} \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

- Consider y' of a particle in an arbitrary other boosted frame F' (e.g. in frame of another moving particle F') vs rest frame using

$$\begin{aligned} p'_{||} &= \gamma (p_{||} + \beta E) \\ E' &= \gamma (E + \beta p_{||}) \end{aligned}$$

$$y' = \frac{1}{2} \ln \left(\frac{E' + p'_{||}}{E' - p'_{||}} \right) = \frac{1}{2} \ln \left(\frac{\cancel{\gamma} (E + p) + \beta (E + p)}{\cancel{\gamma} (E - p) - \beta (E - p)} \right)$$

$$= \frac{1}{2} \ln \left[\left(\frac{E + p_{||}}{E - p_{||}} \right) \left(\frac{1 + \beta}{1 - \beta} \right) \right] = \boxed{y + y_{\beta_F}}$$

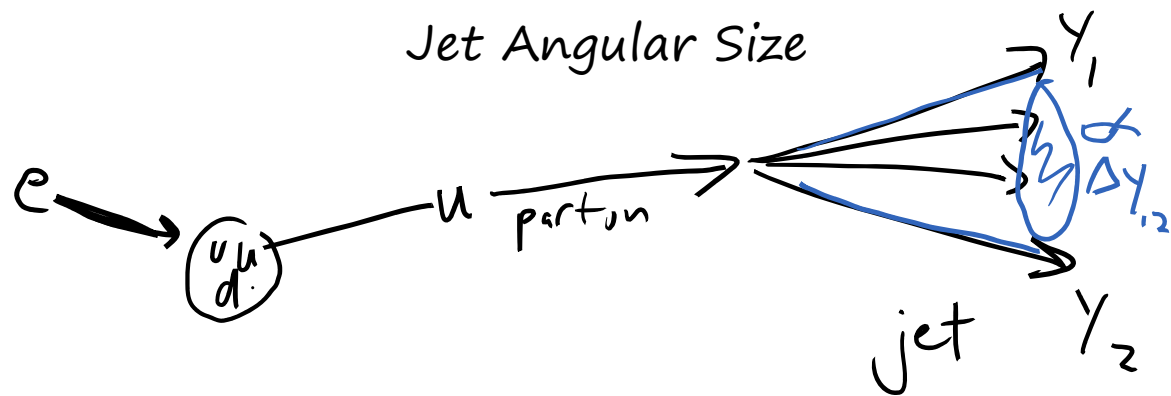
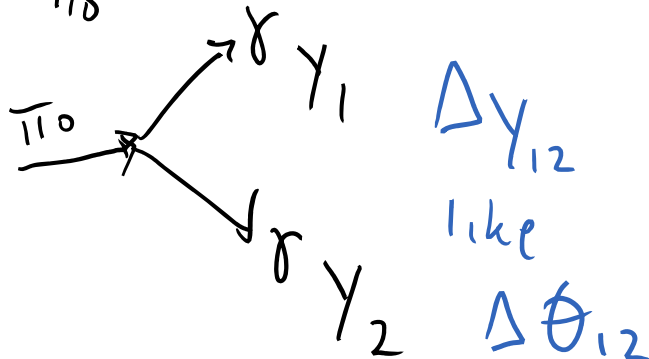
- Thus boosting rapidity just adds a constant equal to y_{β_F}
 - (dependent on the beam only, same for any particle being transformed into that frame)
- Therefore differences in rapidity in one frame (e.g. between two particles) $\Delta y = y_2 - y_1$ are invariant
- This leads us to first reason why it is useful...

Invariance of Δy

- The significance of $\Delta y = y_1 - y_2$ being invariant means that any (usually angular) “size” scale we define, if expressed in this variable will stay invariant in any frame.
- Furthermore we will show that the numerical size of y and η can be interpreted as the angular size *in radians*.
 - i.e. a Δy of 1, the difference $y_2 - y_1$ of two particles rapidities equaling 1, can be interpreted as them having an angular difference of 1 in their unboosted frame (corresponding to the COM frame/lab frame for colliders) for most cases
- Examples:

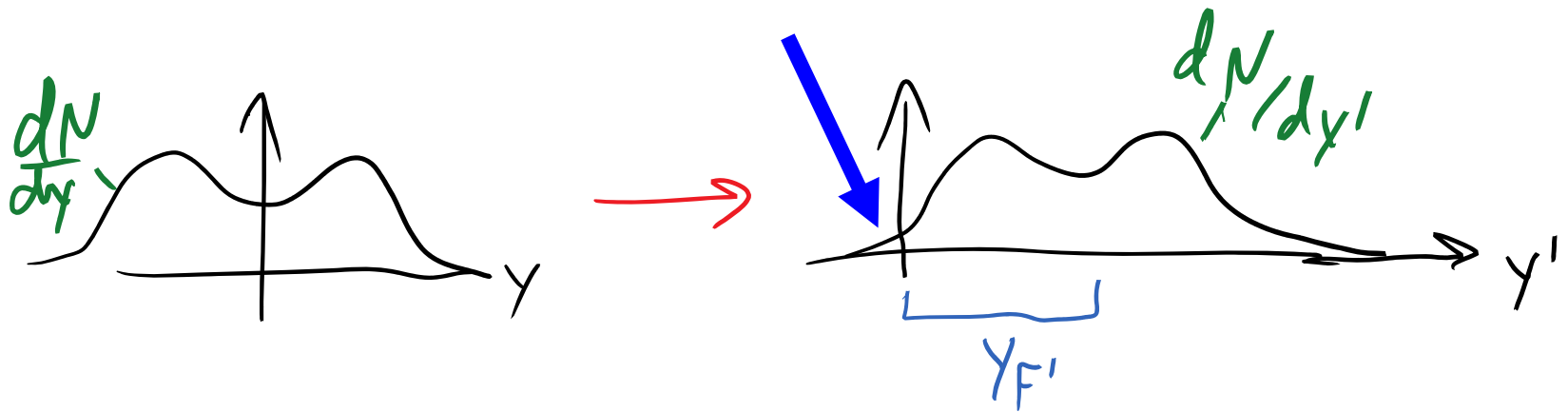
π^0 Decay

$$P_{\pi^0} \Rightarrow \pi^0 \rightarrow \gamma + \gamma$$



dy also Invariant

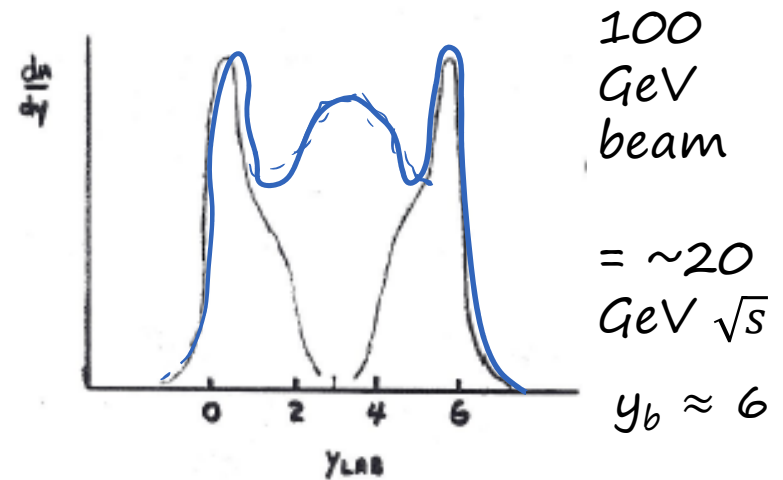
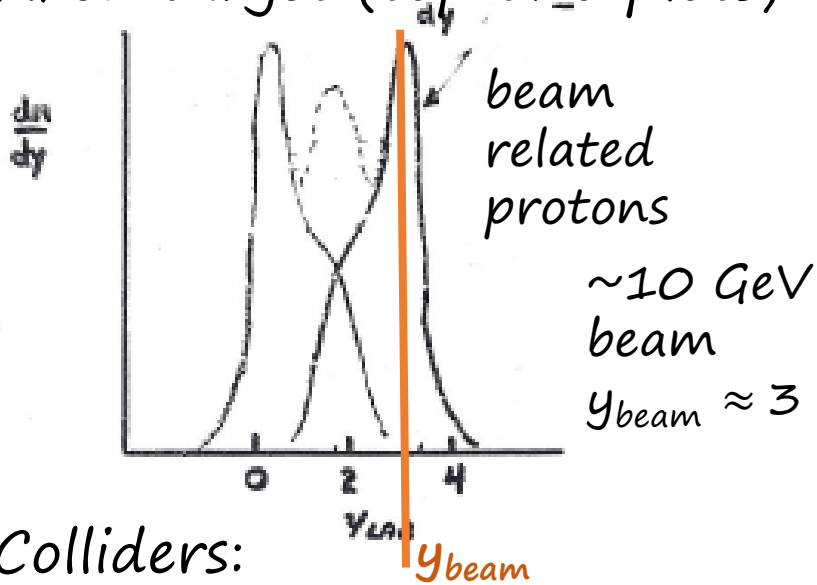
- Amongst many important properties of rapidity the invariance of Δy 's is probably the most important
- Taking the limit $\Delta y = 0 \rightarrow dy$ is also invariant
- This means that (whole) distributions as a function of rapidity ie dN/dy will have the **exact same shape** in boosted frames, just shifted in y by a constant value shift (y_F)



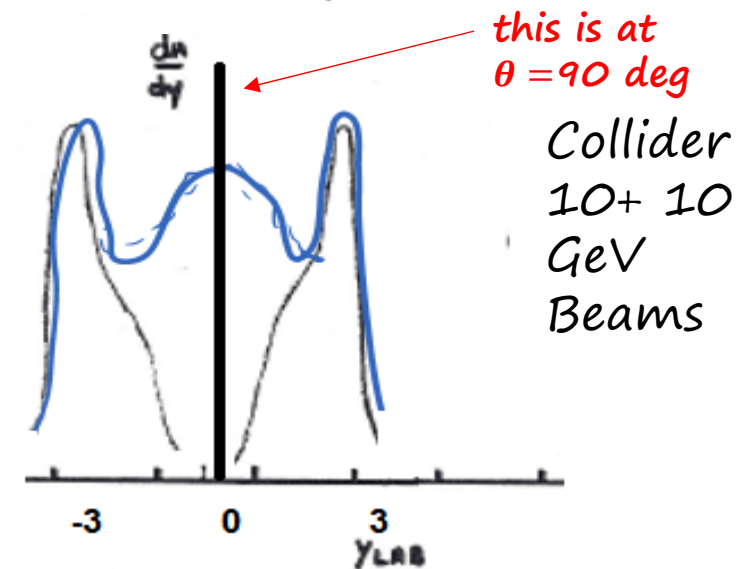
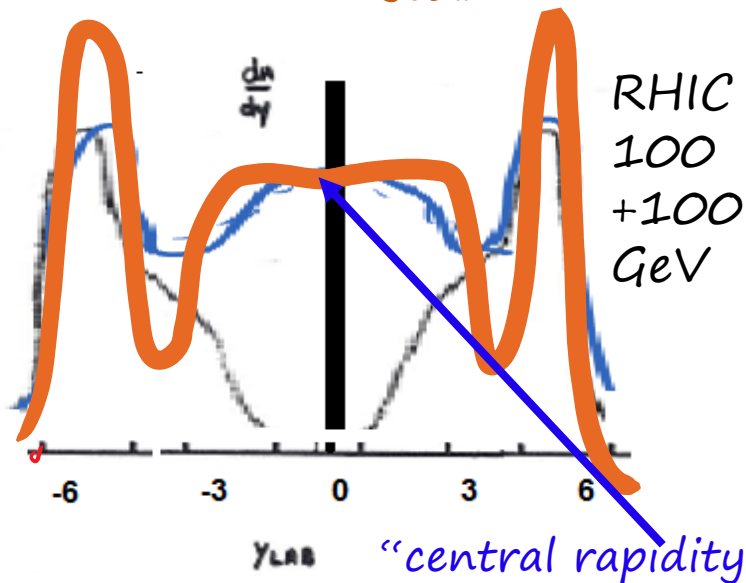
- Ideal e.g. for comparing fixed target vs collider results e.g.
- Recall that y_F in that case would be the beam particle rapidity
- **PRACTICALLY the BEAM PARTICLE RAPIDITY ALSO DEFINES THE LIMIT OF INTERESTING y 's (and therefore η 's) that are interesting.**

Examples of Particle y Distributions

- Hadron distributions for proton beams (& targets):
- Fixed target (top two plots):



- Colliders:

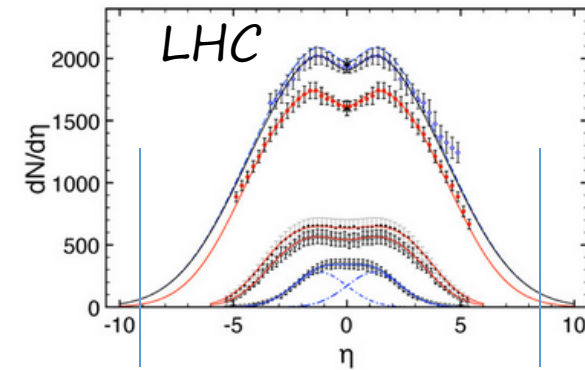
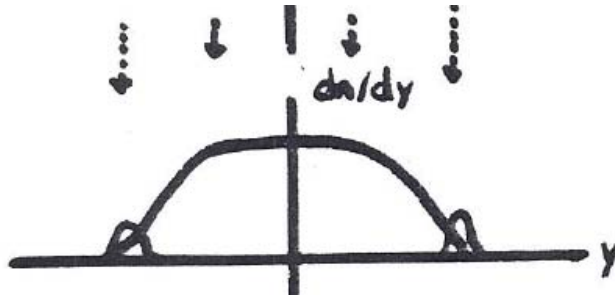


Things to notice about previous slide plots

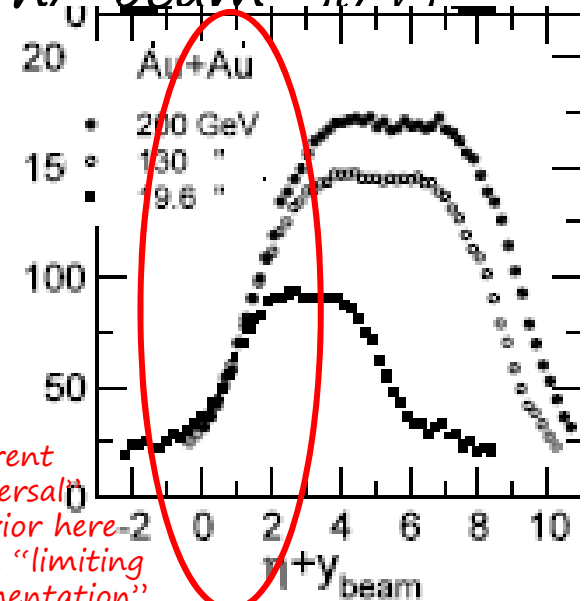
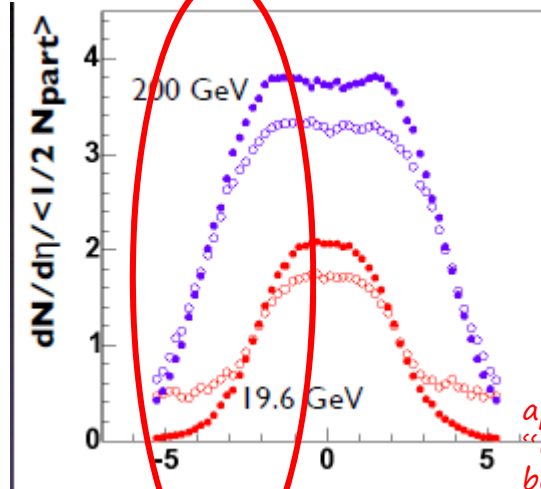
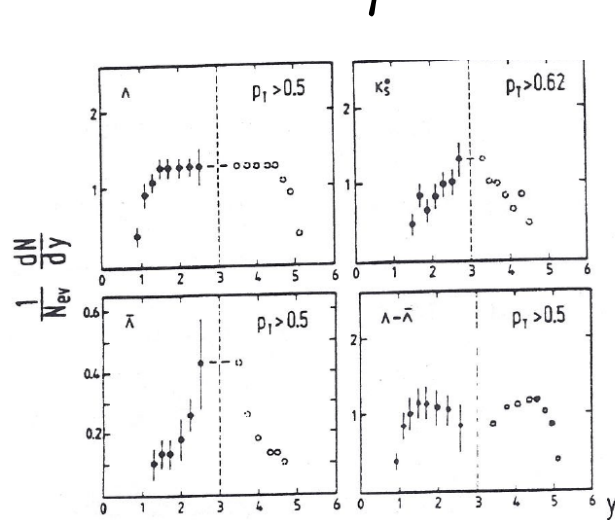
- Each beam energy defines a kind of maximum rapidity value (corresponding to $\theta = \sim 0$ deg wrt beam) and ~all other particles in event have y values (y -angles) lower than the maximum ($y_{\text{beam}} = 3$: 10 GeV; 6 100 GeV; (~ 9 -5 TeV beams at LHC))
 - This makes sense thinking of y as “angle” because angular space get boosted more and more the more p_z the particle has
 - Therefore we can fit more angular ranges in when more energy (and potential higher and higher boosts) available for far forward particles
 - Also as beam energy increases, y -value of dN/dy (dN/dy itself) at center doesn't decrease, distribution mostly just gets wider $\rightarrow$ Extra energy goes into producing more particles to fill up the newly available phase space.
- Comparing **top right** to **bottom right** cases (in green box):, shape of dN/dy is exactly the same, just shifted left. -- in collider case must be symmetric around $y=0 \rightarrow$ this implies fixed target must also be similarly symmetric though not around $y=0$ (instead around $y \approx \frac{1}{2} y_{\text{beam}}$)
 - implying there must be some “activity” around direction “OPPOSITE” to beam in fixed target – note though, this “leftside” peak in that frame does not occur 180 deg from beam direction in fixed target frame.
 - both cases have same total Δy range because they have the same $\sqrt{s}$
 - higher beam energy/beam rap = 6 for one beam in fixed target case (100 GeV) than both beams in collider case (10 GeV+10, $y=3, -3$ —both 3s same as top left fixed target single 10 GeV beam
- Bottom left collider 100 + 100 (look at this last). Double the total Δy ($[-6, 6] \rightarrow 12$ units of rapidity, the $|6|$ is same as for single 100 GeV fixed target beam in top right

Some more typical dN/dy 's , $dN/d\eta$'s

- More realistic even for e.g. protons



- For most particles (not produced pseudoelastically by beam or beam particles themselves) ~ (ie not $p/n/$ "beam" π/v)



apparent
"universal"
behavior here -
called "limiting
fragmentation"

Back to Pseudo-rapidity

- Very closely related quantity : η -- Pseudo rapidity

In experimental particle physics, **pseudorapidity**, a coordinate describing the angle of a particle relative to the beam axis as

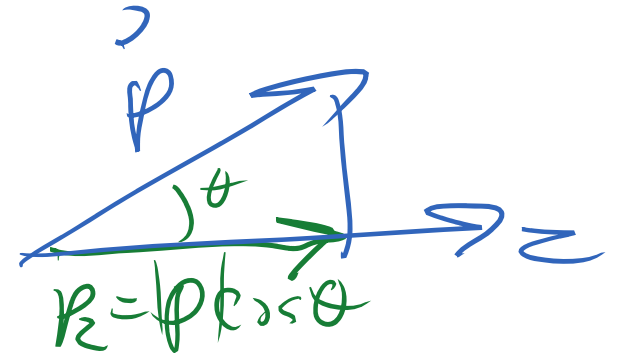
$$\eta \equiv -\ln \left[\tan \left(\frac{\theta}{2} \right) \right],$$

where θ is the angle between the particle three-momentum and the direction of the beam axis.^[1] Inversely,

$$\theta = 2 \arctan(e^{-\eta}).$$

As a function of three-momentum $\mathbf{p}$, pseudorapidity

$$\eta = \frac{1}{2} \ln \left(\frac{|\mathbf{p}| + p_L}{|\mathbf{p}| - p_L} \right) = \operatorname{artanh} \left(\frac{p_L}{|\mathbf{p}|} \right),$$



$$\eta = \frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$$

- Because of its more direct relation to θ ie GEOMETRY of the detector usually (independent of any particle/momenta) it is probably used in MORE situations

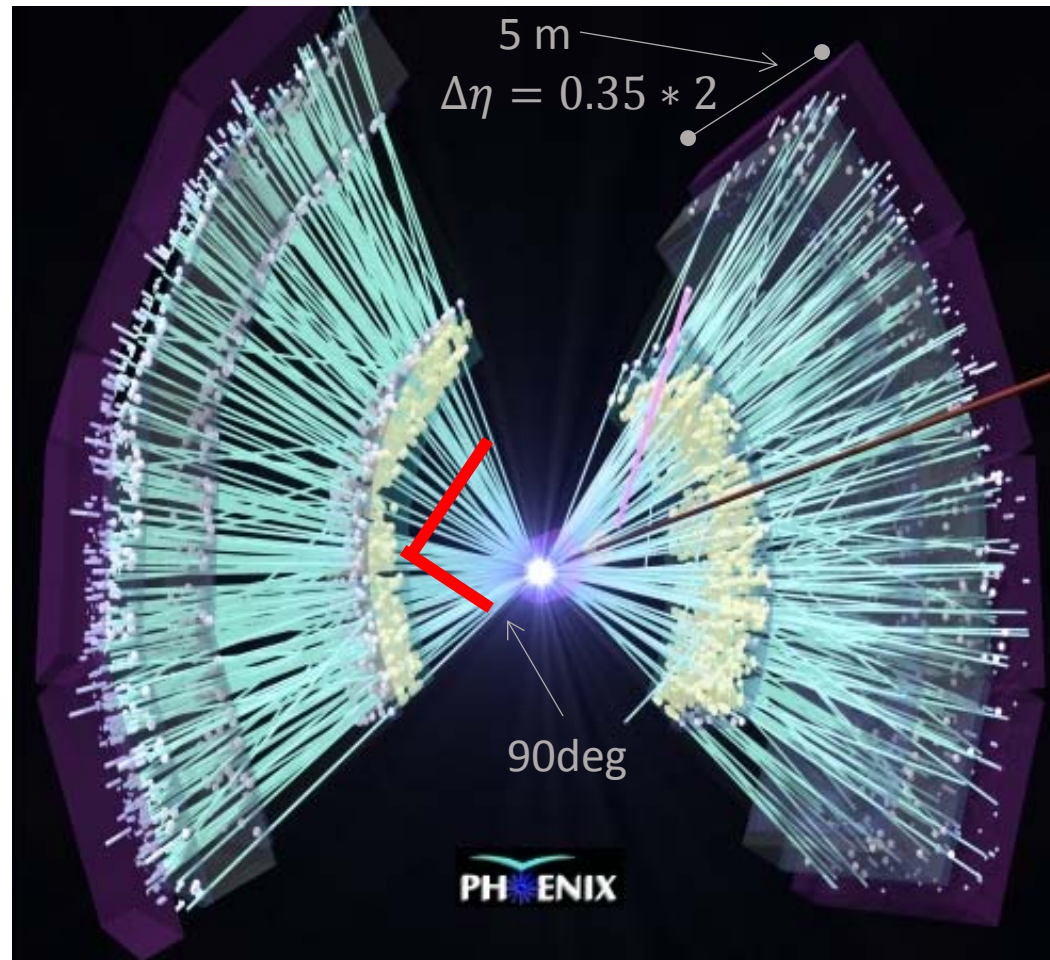
Solid angle & High Energy Hadron-Hadron Collision

- Example : PHENIX “Solid angle”
- Quoted in terms of η
- instead of $\Delta\theta$ using boost invariant $\Delta\eta \sim 2 * 0.35 * \pi$ ($\Delta\phi$)
- sPHENIX will improve rapidity acceptance to $[-1, 1]$ (2/0.7 times better) (and in ϕ π to 2π)

~ 2 sterRadian?

$\sim 1/6$ of 4π

- From areas : faces are 5 m from beam point and 5 m along beam
 - 1 arm ~ 7 m arc length 5m along beam
 - “rectangles of $5 \times 7 \text{ m} = 35 \text{ m}^2$ ”
 - $35 \text{ m} / 4\pi (5 \text{ m})^2$
 $= A_{\text{phx}} / A_{\text{full sphere}}$
 $= 2 * 35 / 270 \sim 2/9 \sim 1/5$
 fraction of total 4π solid angle a little different but close to $1/6$ above
 $\Delta\Omega = 4\pi * (1/5) = \sim 2 \text{ sR}$



$$|d\eta| = |d\theta|$$

- From the definition of η it is easy to show that at 90 degrees (at $\eta = y = 0$) $d\eta = d\theta$:

$$\eta = -\frac{1}{2} \ln \left(\tanh \frac{\theta}{2} \right)$$

$$\frac{d\eta}{d\theta} = -\frac{1}{2} \frac{1}{\tanh \theta/2} \left(\frac{1}{\cosh^2 \theta/2} \right)$$

$\downarrow \quad \theta = 90^\circ \quad \downarrow$
 $(\frac{\theta}{2} = 45^\circ)$

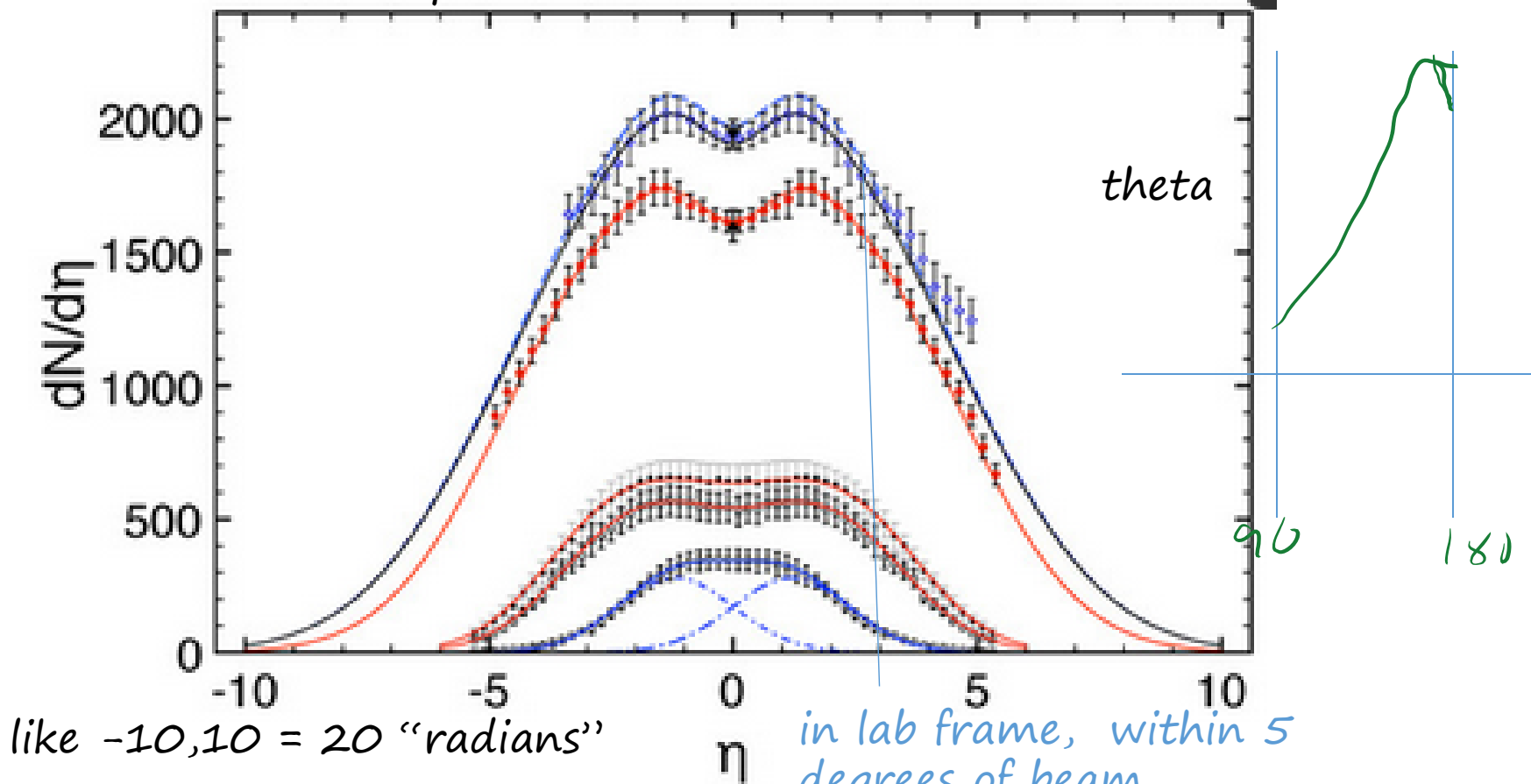
$$d\eta = -d\theta$$

$= 1 / 1/2$

- This means the scale of $d\eta$ or $\Delta\eta$ is in radians (there @ 90)
- By the boost invariance of $d\eta = dy$, this means its scale is equivalent to radians at ALL y !!!!

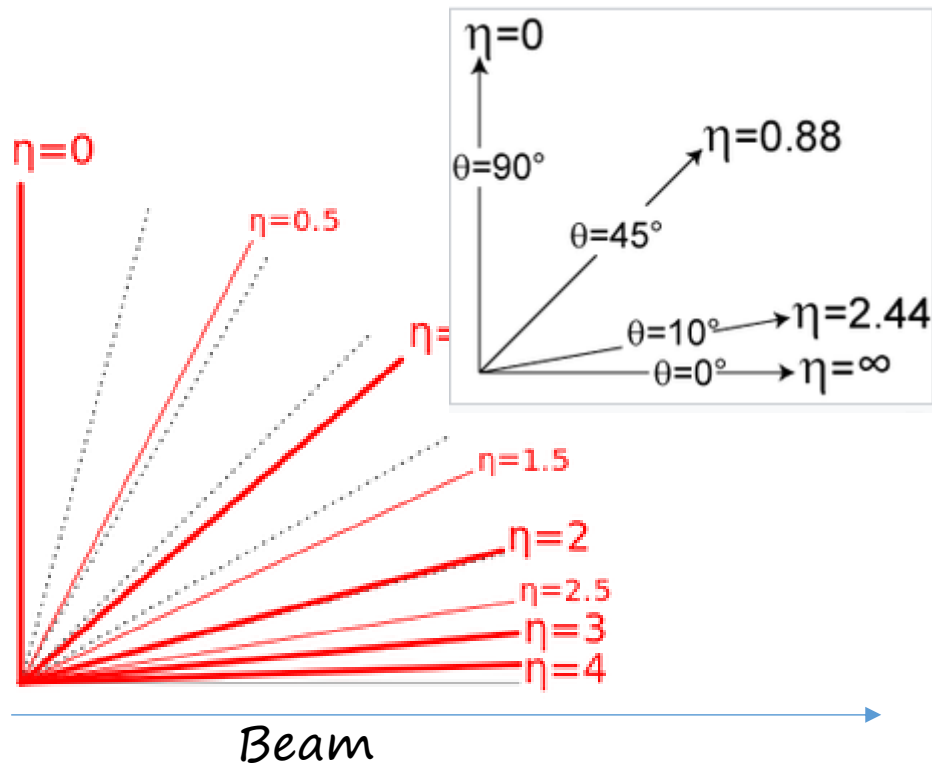
Implications : More Space

- We are fitting more radians in with greater energy!!!
- We have “more space” to work with because of boosts!
- More than anything greater than ~ 10 TeV collider beam energies $y_b = [-9, 9]$ has nearly 6π (20 rad) angular space in the longitudinal (beam) direction!
- Generally more energy more particles – where – good way to remember? $\sim$ Isotropic in boosted frames



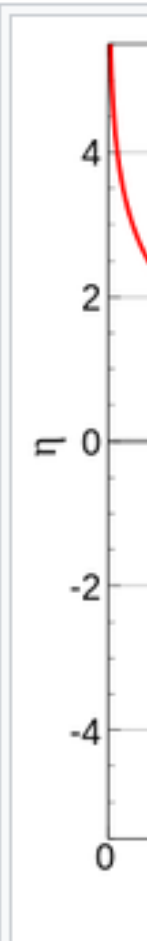
Numerical examples $\eta, y \leftrightarrow \theta$

- Pseudo rapidity allows an easier mapping to θ and shows us why this version of θ is so much better
- (From Wikipedia:pseudorapidity):



representative values:

θ	η	θ	η
0°	∞	180°	$-\infty$
0.1°	7.04	179.9°	-7.04
0.5°	5.43	179.5°	-5.43
1°	4.74	179°	-4.74
2°	4.05	178°	-4.05
5°	3.13	175°	-3.13
10°	2.44	170°	-2.44
20°	1.74	160°	-1.74
150°	-1.32		



Other things about η : Jets

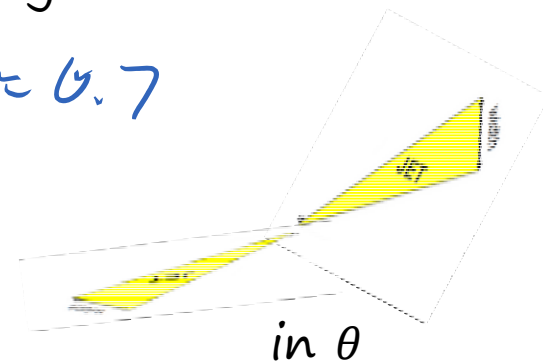
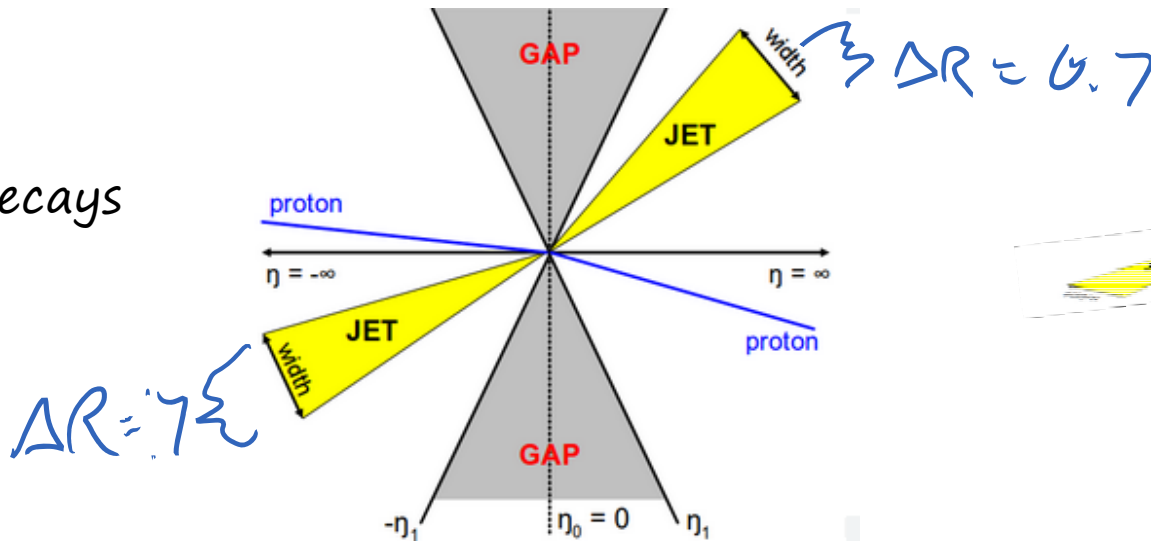
- Useful things mentionned from looking at wiki page

$$\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$$

and “ $\vec{R}$ ” = $[\eta, \phi]$ is used somewhat like solid angle -- instead of solid angle in fact acceptance is given in terms of “ $\vec{DR}$ ”, $\Delta\phi$, Delta eta w/ (respect to beam center of detector) and elements of detector are identified by their η, ϕ locations

-R is used a lot when talking about QCD jets $\rightarrow$ jet angular size
 $\Delta R \sim 0.7$

-Jets and Decays



η Hyperbolic Relations

- It's best to use these hyperbolic relations to use η to determine things like p_T

$$p_z = p_T \sinh \eta, \quad |\mathbf{p}| = p_T \cosh \eta. \quad (\cos \theta = \tanh \eta !)$$

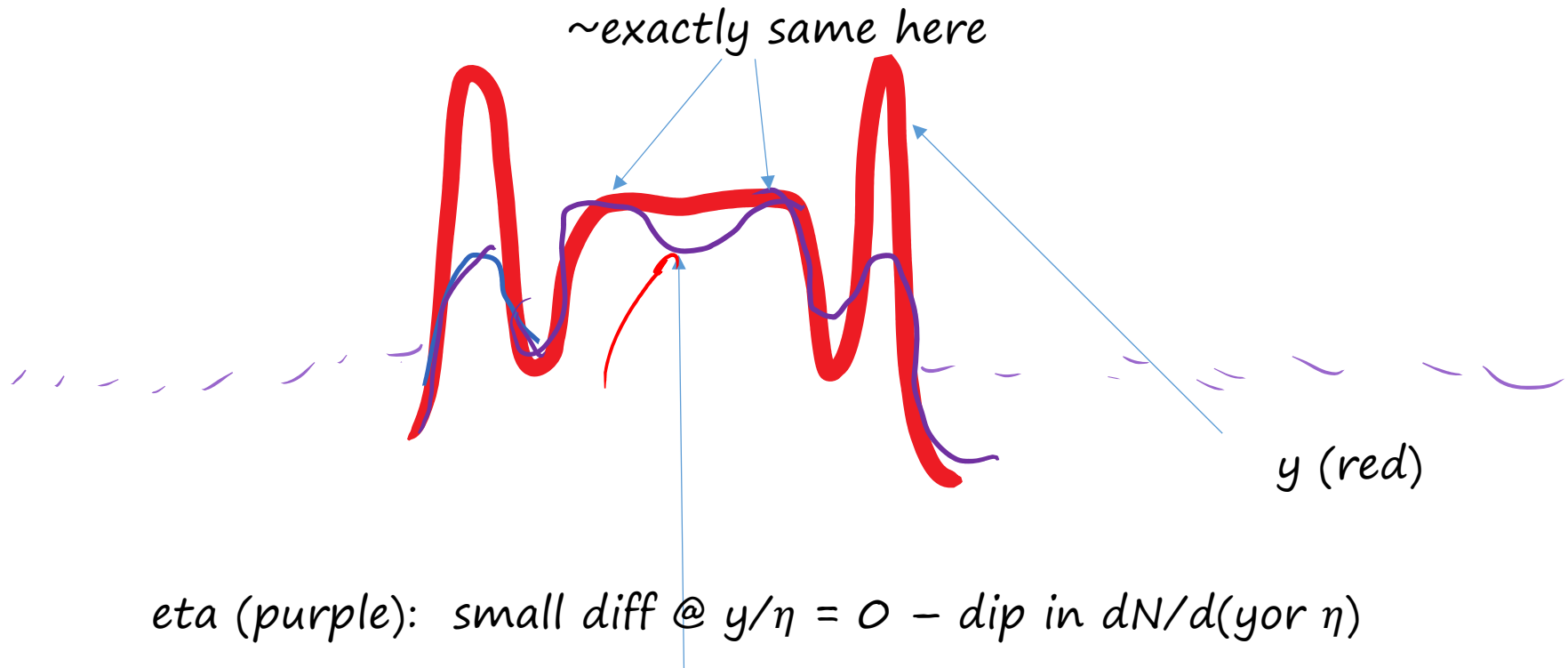
- (Note normally:

- $p_T = p \sin \theta$
- $p_z = p \cos \theta$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

Pseudorapidity η vs Rapidity y

- Differences from y vs η



Obviously “MAX” pseudo rapidity defined by “max” rapidity ie beam rapidity (not really max but only e.g. decays will go higher in practice) since beam pseudo-rap $\rightarrow \ln(p - p_z) = \ln(p_z - p_z) = \ln(0) = \infty$

Will explore in problem set – wait for some clarifications

Cosh, Sinh, Tanh y – The other rapidity

- Obviously $e^{2y} = (1+\beta)/(1-\beta)$ $y = \frac{1}{2} \ln \left(\frac{1+\beta}{1-\beta} \right)$ $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$
- It shouldn't be too surprising that $\tanh y = \beta$ (if F' or else, = " $\beta_{||}$ ")
 - $f(x) = 1-x/(1+x)$ is inverse of itself (over $[0,1]$)—"involution" (nearly so elsewhere)
 - this fact is useful for some other cases in particle physics especially "asymmetries" $\rightarrow (x_1-x_2)/(x_1+x_2)$
- Further, for the situation 1 case, : $\gamma = \cosh y$ and $\gamma\beta = \sinh y$ – convenient e.g. for coding
- For this reason, sometimes rapidity is first introduced as the "hyperbolic angle" of a boost—like an alt way of specifying β and γ .

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh w & -\sinh w \\ -\sinh w & \cosh w \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} = \Lambda(w) \begin{pmatrix} ct \\ x \end{pmatrix} .$$

- But note this is a different kind of rapidity – often a different symbol is used like w or ξ –situation 1, but used in a different way

• ...

The other rapidity: not as useful

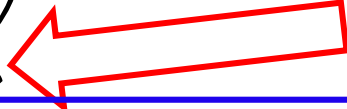
- Obviously $e^{2y} = (1+\beta)/(1-\beta)$ $y = \frac{1}{2} \ln \left(\frac{1+\beta}{1-\beta} \right)$ $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$
- It shouldn't be too surprising that $\tanh y = \beta$ (if F' or else, = " $\beta_{||}$ ")
- Further, for the situation 1 case, : $\gamma = \cosh y$ and $\gamma\beta = \sinh y$
- But note this is a different kind of rapidity – often a different symbol is used like w or ξ –situation 1, but used in a different way
- ...because it gives the impression that you might define different “rapidities” for each particle produced in a collision in the situation 1 way, for ALL particles ie as boosts in each particles direction



- but these “rapidities” in different directions would not have the useful transformation property (invariance of Δy 's)
 - i.e. don't think of y as boost “angle”/ boost amount for each particle
- Nonetheless the $\tanh y = \beta_{||}$ formula (situation 2) is still useful sometimes

Reminder: Particle Physics Rapidity

- The key difference btw the “other” and the “regular/usual/particle physics” use of rapidity are obviously the directions w.r.t “l”
- Experimentally, for the “usual” kind, we typically **ONLY** consider boosts (& l direction as being) along the (common) beam direction p_z
-- **common to all particles**
 - where as the other is the in each particles direction
- So for us

$$y = \frac{1}{2} \ln \left(\frac{E + p_{z/\text{beam}}}{E - p_{z/\text{beam}}} \right)$$


- The “OTHER” kind of rapidity (situation 1 used for ALL particles, not just beam particles) seems to me like something used in derivations e.g. by theorist sometimes but not ever in relation to experimental data
- For anything even loosely related to experimental data, the “beam direction boost” rapidity (situation 2, with F chosen as the z/beam direction) is always what is meant—according to the above formula.
- Key thing then: all particles $y' = y + y_{\text{beam}} \rightarrow \Delta y = y_2 - y_1 = \Delta y'$ any pair
 - because same boost y_{beam} for all particles